

**ĐẠI HỌC QUỐC GIA HÀ NỘI
TRƯỜNG ĐẠI HỌC CÔNG NGHỆ**



Nguyễn Thành Phát

CAUSAL INFERENCE ON TIME SERIES DATA

KHOÁ LUẬN TỐT NGHIỆP

Ngành: Khoa học máy tính

HÀ NỘI - 2026

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UNIVERSITY OF ENGINEERING AND TECHNOLOGY**



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CAUSAL INFERENCE ON TIME SERIES DATA

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Lời cảm ơn

Lời đầu tiên, tôi xin gửi lời cảm ơn chân thành và lòng biết ơn sâu sắc tới thầy TS. Nguyễn Văn Sơn, người đã tận tình hướng dẫn tôi xuyên suốt thời gian thực hiện khóa luận tốt nghiệp này.

Tôi xin cảm ơn đội ngũ CrunchDAO nói chung và anh Enzo Caceres nói riêng đã tạo ra một nền tảng, sân chơi và hỗ trợ tôi xuyên suốt quá trình thực hiện khóa luận tốt nghiệp này.

Tôi cũng xin gửi lời cảm ơn chân thành tới các thầy, cô hiện đang công tác và giảng dạy tại trường Đại học Công Nghệ - Đại học Quốc gia Hà Nội đã tạo mọi điều kiện thuận lợi cho tôi học tập và nghiên cứu.

Hà Nội, ngày tháng năm

Sinh viên

Nguyễn Thành
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Lời cam đoan

Tôi là Nguyễn Thành Phát, sinh viên lớp K67I-CS3. Tôi xin cam đoan khóa luận *"Suy luận nhân quả trên dữ liệu chuỗi thời gian"* là công trình nghiên cứu do bản thân tôi thực hiện. Các nội dung nghiên cứu, kết quả trong khoá luận là xác thực.

Trong quá trình thực hiện khóa luận, tôi có sử dụng các công cụ Trí tuệ nhân tạo (AI) như một phương tiện hỗ trợ trong việc: rà soát lỗi cú pháp, tối ưu hóa phong cách viết trong báo cáo, và hỗ trợ chuyển đổi định dạng tài liệu. Tôi cam đoan rằng các công cụ này chỉ đóng vai trò hỗ trợ kỹ thuật.

Các thông tin sử dụng trong khoá luận là có cơ sở và không có nội dung nào sao chép từ các tài liệu mà không ghi rõ trích dẫn tham khảo. Tôi xin chịu trách nhiệm về lời cam đoan này.

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Hà Nội, ngày tháng năm

Sinh viên

Nguyễn Thành
Phát

Tóm tắt

Suy luận nhân quả trên dữ liệu chuỗi thời gian là một lĩnh vực then chốt trong khoa học dữ liệu, đóng vai trò quan trọng trong việc hiểu rõ cơ chế vận hành của các hệ thống phức tạp thay vì chỉ dừng lại ở phân tích tương quan thống kê [1]. Trong báo cáo này, tôi trình bày nghiên cứu về bài toán xác định cấu trúc đồ thị có hướng không chu trình (Directed Acyclic Graph - DAG) từ dữ liệu chuỗi thời gian đa biến dựa trên bối cảnh cuộc thi do ADIA Lab phối hợp cùng CrunchDAO tổ chức.

Bài toán tập trung vào việc xác định mối quan hệ nhân quả giữa hai biến đặc biệt: biến can thiệp (X) và biến kết quả (Y), đồng thời phân loại vai trò của các biến còn lại trong hệ thống. Các biến này được phân định vào tám danh mục chức năng cụ thể bao gồm: biến gây nhiễu (Confounder), biến điều tiết (Collider), biến trung gian (Mediator), biến độc lập, và các biến thuộc chuỗi hệ quả hoặc nguyên nhân của X và Y [1]. Đây là một thách thức lớn do sự hiện diện của các yếu tố nhiễu và cấu trúc phụ thuộc phức tạp trong dữ liệu thực tế.

Để giải quyết bài toán này, tôi khảo sát và áp dụng các phương pháp suy luận nhân quả hiện đại, bao gồm cả hướng tiếp cận không giám sát (unsupervised) dựa trên kiểm định tính độc lập điều kiện và hướng tiếp cận có giám sát (supervised) tận dụng sức mạnh của các mô hình học máy. Quá trình thực nghiệm được thực hiện trên tập dữ liệu quy mô lớn gồm 47,000 mẫu được cung cấp bởi nền tảng CrunchDAO [1], với số lượng biến biến thiên nhằm kiểm chứng tính bền vững và khả năng tổng quát hóa của các mô hình.

Kết quả nghiên cứu được đánh giá dựa trên chỉ số độ chính xác cân bằng đa lớp (Multiclass Balanced Accuracy), tập trung vào khả năng nhận diện chính xác vai trò logic của các nút mạng đối với cặp biến $X \rightarrow Y$. Nghiên cứu không chỉ dừng lại ở việc dự đoán sự tồn tại của các cạnh mà còn hướng tới việc tinh chỉnh mô hình để giảm thiểu sai số trong việc xác định cấu trúc hệ thống, từ đó cung cấp một quy trình hiệu quả cho việc khám phá tri thức từ dữ liệu chuỗi thời gian.

Từ khóa: Suy luận nhân quả (Causal Discovery), Chuỗi thời gian (Time Series), Đồ thị có hướng không chu trình (DAG), CrunchDAO [1].

Abstract

Causal discovery on time series data is a pivotal field in data science, playing a crucial role in understanding the underlying mechanisms of complex systems rather than merely relying on statistical correlation analysis [1]. In this report, I present a study on the problem of discovering Directed Acyclic Graph (DAG) structures from multivariate time series data, based on the competition organized by ADIA Lab in collaboration with CrunchDAO.

The problem focuses on identifying the causal relationship between two primary variables: the treatment (X) and the outcome (Y), while simultaneously classifying the roles of the remaining variables within the system. These variables are categorized into eight functional roles, including: confounders, colliders, mediators, independent variables, and variables belonging to the causal or consequential chains of X and Y [1]. This task presents a significant challenge due to the presence of noise and the complex dependency structures inherent in real-world time series data.

To address this problem, I investigate and apply modern causal discovery methods, encompassing both unsupervised approaches based on conditional independence testing and supervised approaches leveraging the power of machine learning models. The experimental process was conducted on a large-scale dataset comprising 47,000 samples provided by the CrunchDAO platform [1], with varying numbers of variables to verify the robustness and generalization capabilities of the models.

Research results were evaluated using the Multiclass Balanced Accuracy metric, focusing on the ability to accurately identify the logical roles of nodes relative to the $X \rightarrow Y$ pair. This study goes beyond merely predicting the existence of edges, aiming to refine models to minimize errors in identifying the system's logical structure, thereby providing an effective framework for discovering causal knowledge from time series data.

Keywords: Causal Discovery, Time Series, Directed Acyclic Graph (DAG), CrunchDAO [1].

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Danh mục các từ viết tắt

Từ viết tắt	Cụm từ đầy đủ	Ý nghĩa / Mô tả
BiLSTM	Bidirectional Long Short Term Memory	Mô hình bộ nhớ dài hạn ngắn hạn hai chiều
BO	Buffer Overflow	Lỗi tràn bộ đệm

Chapter 1

Introduction

1.1 Context and Problem Statement

Understanding why things happen matters more than merely knowing that things happen together. When a doctor prescribes a treatment, an economist evaluates a policy, or an engineer diagnoses a system failure, the goal is not to find correlations but to understand causal mechanisms [2]. Knowing that treatment and recovery are correlated is insufficient for making good medical decisions; what matters is whether the treatment causes recovery or whether both are confounded by some third factor such as patient age or disease severity.

This thesis addresses the **ADIA Lab Causal Discovery Challenge**, a competitive machine learning benchmark organized by ADIA Lab in collaboration with CrunchDAO [1, 3]. The task requires classifying the causal role of each variable in a dataset with a designated treatment X and outcome Y . Given $n = 1000$ observational samples over $p \in \{3, \dots, 10\}$ variables, the model must assign each remaining variable K to one of eight causal roles: Confounder, Mediator, Collider, Independent, Cause of X , Consequence of X , Cause of Y , or Consequence of Y .

This eight-class classification problem presents four specific challenges that classical causal discovery methods struggle to overcome. First, subtle class distinctions make role pairs difficult to separate: Confounder ($K \rightarrow X, K \rightarrow Y$) and Cause of X ($K \rightarrow X$ only) both induce correlation between K and X ; similarly, Mediator ($X \rightarrow K \rightarrow Y$) and Consequence of X ($X \rightarrow K$ only) both exhibit dependence between X and K . The distinguishing signal lies in subtle asymmetries that scalar statistics cannot capture. Second, class imbalance causes models to neglect minority roles such as Mediator and Collider without careful handling strategies. Third, permutation invariance requires the model to treat (X, Y, K_1, K_2) identically to (K_2, X, K_1, Y) since variable naming order carries no semantic information. Fourth, nonlinear relationships mean that simple Pearson correlation is often insufficient when causal roles differ in the shape rather than strength of their conditional

relationships.

1.2 Existing Solutions and Their Limitations

Prior work on this challenge falls into three groups, each with fundamental limitations that constrain performance.

Classical Causal Discovery Methods rely on conditional independence tests such as the PC algorithm to identify causal structure from observational data [4]. These methods achieve only **40% Balanced Accuracy** on the ADIA Lab challenge [5], barely above the 12.5% random baseline for eight-class classification. Their limitations are severe: ****limited to linear relationships and simple dependencies****, and ****cannot handle the eight-class fine-grained classification task**** since their output is a causal graph, not a role assignment for each variable.

Machine Learning Feature Engineering Approaches extract scalar statistics from each variable pair and train gradient boosted classifiers. A LightGBM model on 11 hand-crafted features achieves 63.11%. A comprehensive pipeline with 300+ features including Pearson, Spearman, and distance correlation coefficients, partial correlations, conditional mutual information, HSIC, ANM scores, and outputs from PC, LiNGAM, NOTEARS, and GES algorithms reaches 72.64% with XGBoost and LightGBM ensembles. Despite this extensive feature engineering, performance remains 8.44% below the final model. The core problem is that ****scalar features irreversibly discard curve shape information****: conditional expectation curves encode the functional form of causal relationships, and reducing them to single numbers loses the distinguishing patterns between roles.

Deep Learning Approaches began with the competition’s first-place solution, which introduced 3-channel edge tensors encoding conditional distributions as structured inputs for a neural classifier [6]. This approach achieves 76.70% Balanced Accuracy, establishing a strong baseline that outperforms all classical and ML feature engineering methods. However, it has three critical limitations: ****only 3 channels miss multi-scale patterns**** that multi-bandwidth kernel regression can capture; ****no structural awareness in the attention mechanism**** means the model cannot encode topology-aware priors for chain, fork, and collider relationships; and ****no 2D visual representation of joint distributions**** leaves patterns invisible to

pairwise analysis.

These limitations collectively define the performance ceiling of existing approaches. Addressing all eight requires a fundamentally different architecture that preserves curve shape information, captures multi-scale causal asymmetry, encodes structural inductive bias, and combines complementary visual and statistical representations.

1.3 Our Solution and Results

This thesis proposes a dual-pipeline neural architecture that addresses all eight limitations through five targeted contributions. The first pipeline processes 1D edge tensors encoding conditional distributions; the second processes 2D scatter density images encoding joint distribution topology. These pipelines are fused through learned gating and trained with a weighted dual-loss objective.

The final model achieves **81.082% Balanced Accuracy** on the private test set, surpassing the first-place solution by +4.38 percentage points and outperforming the faithful baseline reimplementation by +7.12 percentage points.

1.4 Five Technical Contributions

The first contribution is an **8-Channel Edge Tensor** that extends the baseline 3-channel representation by adding multi-bandwidth kernel regression channels ($h \in \{0.2, 0.5, 1.0\}$) capturing causal asymmetry at different scales, and ANM residual channels providing directional asymmetry signals. This representation captures patterns that a single bandwidth cannot, yielding +1.89% improvement over the 3-channel baseline.

The second contribution is a **Structural Attention Bias** that injects topology-aware priors into self-attention layers. Rather than adding thousands of unconstrained parameters (which produced negative gains of -1.19% to -1.90% in failed experiments), this approach encodes six causal relationship types using only six learned scalars, outperforming those larger models by over 3.5 percentage points and providing the largest single architectural gain of +2.65%.

The third contribution is a **2D Scatter Density Pipeline** that renders joint distributions as 8-channel 32×32 density images processed by a convolutional

encoder. This complements the 1D edge tensor pipeline by capturing patterns such as multimodality and conditional heteroskedasticity that are invisible to pairwise scalar statistics, contributing +0.88% with particularly strong gains on Confounder (+2.78%) and Consequence of Y (+2.18%).

The fourth contribution is **X/Y Remap Data Augmentation** that treats each directed edge $A \rightarrow B$ in the ground-truth DAG as a new treatment-outcome pair (X', Y') , recomputing all eight role labels accordingly. This multiplies effective training data by approximately 11x without additional data collection, yielding +1.55% to +3.32% depending on base model configuration.

The fifth contribution is a **Dual-Head Loss Weighting** scheme with $\lambda = 0.7$ balancing the edge and node loss heads. This ensures both pipelines remain causally meaningful throughout training rather than one dominating, contributing an additional +0.59% improvement.

Table 1.1: Performance comparison on the ADIA Lab Causal Discovery Challenge. Classical methods suffer from linear assumptions and coarse-grained output. ML feature engineering irreversibly loses curve shape when compressing to scalars. The 3-channel deep learning baseline misses multi-scale patterns, lacks structural awareness, and has no visual representation. The proposed method addresses all eight limitations simultaneously, achieving state-of-the-art performance.

Approach	Balanced Accuracy	Gap to SOTA
Classical baselines (PC, RF, DL) [5]	$\sim 40\%$	-41%
ML fullstack ensemble (300+ features)	72.64%	-8.44%
Competition 1st place (3-channel) [6]	76.70%	-4.38%
Proposed (8-channel + 2D + XY aug)	81.082%	Baseline

1.5 Validation Evidence

Ablation studies confirm that each contribution is necessary rather than redundant. Removing the edge context encoder collapses performance from 80.47% to 51.51%, a 29-percentage-point drop demonstrating that the 1D pipeline provides structurally necessary information that the 2D pipeline alone cannot capture. The four directional embeddings ($K \rightarrow X, K \rightarrow Y, X \rightarrow K, Y \rightarrow K$) propagate graph-

wide causal context to each variable’s representation.

The structural attention bias experiments provide a particularly instructive comparison. Adding 256 unconstrained attention parameters produced -1.19% while adding 1024 parameters produced -1.90%. By contrast, the six structured scalars encoding topology priors produced +2.65%. This demonstrates that for this problem, structured inductive bias consistently outperforms raw model capacity when training data represents causal structure rather than arbitrary functions.

Per-class ablation analysis reveals complementary benefits across contributions. The 2D pipeline benefits Confounder (+2.78%) and Mediator (+2.28%) most, roles where joint distribution shape provides distinguishing information. The XY augmentation benefits rare classes most, validating that data multiplication addresses class imbalance rather than just overall volume.

1.6 Thesis Organization

The remainder of this thesis is organized as follows. Chapter 2 reviews causal graph foundations, formally defines the eight causal roles, explains edge tensor construction and ANM residuals, and describes the deep learning components employed in this work. Chapter 3 presents the complete dual-pipeline architecture including the 8-channel edge tensor, structural attention bias, 2D scatter density classifier, fusion mechanism, and training procedures. Chapter 4 describes the ADIA Lab dataset, evaluation metrics, baseline implementations, and experimental protocols. Chapter 5 reports ablation studies, per-class performance analysis, mechanistic insights, and general lessons extracted from both successful and failed experiments.

Chapter 2

Background and Related Work

This chapter provides the theoretical and methodological foundations for the problem and solution developed in this thesis. It covers causal graph theory and the eight variable roles, the edge tensor representation, deep learning components, existing approaches on the ADIA Lab Challenge, and data augmentation strategies for causal discovery. The presentation emphasizes foundational citations throughout.

2.1 Causal Graph Theory and the Eight Variable Roles

2.1.1 Structural Causal Models and Directed Acyclic Graphs

A structural causal model (SCM) formalizes the relationship between causal structure and observational data [2, 7]. An SCM consists of a set of structural equations, each expressing a variable as a deterministic function of its direct causes plus an independent noise term, together with a directed acyclic graph (DAG) representing the causal parents of each variable. The DAG encodes conditional independence relationships through d -separation, though this section focuses on the graph structure itself rather than the separation criterion [8].

Formally, let $\mathbf{X} \in \mathbb{R}^p$ be a p -dimensional random vector with ground-truth DAG $\mathcal{G} = (\mathbf{V}, \mathbf{E})$ where each variable $V_i \in \mathbf{V}$ has causal parents $\text{PA}(i) = \{V_j : (j, i) \in \mathbf{E}\}$. The structural equation for V_i takes the form $V_i := f_i(\text{PA}(V_i), \varepsilon_i)$ where ε_i is an independent noise variable and f_i is the structural function [9]. The joint distribution $p(\mathbf{X})$ is induced by these equations, and the Markov factorization property states that $p(\mathbf{X}) = \prod_i p(V_i \mid \text{PA}(V_i))$.

In the ADIA Lab Causal Discovery Challenge, the DAG is assumed to contain a distinguished treatment variable X and outcome variable Y connected by the edge $X \rightarrow Y$, with additional variables K that may occupy different causal positions relative to this pair. The task requires classifying each K based on its causal relationship to (X, Y) . The eight possible positions are derived from the adjacency

matrix $\mathbf{A} \in \{0,1\}^{p \times p}$ by examining four binary entries: $\mathbf{A}_{K,X}$ (presence of edge $K \rightarrow X$), $\mathbf{A}_{X,K}$ (presence of edge $X \rightarrow K$), $\mathbf{A}_{K,Y}$, and $\mathbf{A}_{Y,K}$. Table 2.1 enumerates these eight positions and their corresponding edge configurations.

Table 2.1: Label derivation from the adjacency matrix for variable K . Each label is determined by the four binary entries of $\mathbf{A}$ corresponding to edges incident to K with respect to X and Y . The first row for each label shows the four bits in the order $K \rightarrow X$, $X \rightarrow K$, $K \rightarrow Y$, $Y \rightarrow K$.

Label	$K \rightarrow X$	$X \rightarrow K$	$K \rightarrow Y$	$Y \rightarrow K$
Confounder	1	0	1	0
Mediator	0	1	1	0
Collider	0	1	0	1
Cause of X	1	0	0	0
Cause of Y	0	0	1	0
Consequence of X	0	1	0	0
Consequence of Y	0	0	0	1
Independent	0	0	0	0

2.1.2 The Eight Causal Roles

A **Confounder** is a variable K that has causal edges to both X and Y , satisfying $K \rightarrow X$ and $K \rightarrow Y$. This structure makes K a common cause of X and Y , inducing spurious correlation between X and Y when K is not conditioned upon. Confounders are central to Simpson’s paradox and to the problem of confounding bias in causal inference [2]. In the ADIA Lab setting, the presence of an edge $K \rightarrow Y$ distinguishes a Confounder from a Cause of X despite both having $K \rightarrow X$, which is why the scalar statistics that characterize these roles must capture edge-level directionality.

A **Mediator** lies on the causal pathway from X to Y , satisfying $X \rightarrow K \rightarrow Y$. The variable K transmits part of the causal effect from X to Y along a pathway parallel to the direct edge $X \rightarrow Y$ [9]. Mediated effects decompose the total effect of X on Y into direct and indirect components; blocking the mediator at K would interrupt this transmission pathway. The defining feature of a Mediator is that it is both an

effect of X ($X \rightarrow K$) and a cause of Y ($K \rightarrow Y$), creating a chain structure that contrasts with a simple consequence of X where $K \rightarrow Y$ is absent.

A **Collider** is a variable K that has both X and Y as direct causes, satisfying $X \rightarrow K$ and $Y \rightarrow K$. This creates a fork structure in the opposite direction: while a Confounder ($K \rightarrow X, K \rightarrow Y$) induces correlation between X and Y , a Collider induces independence between X and Y in the marginal distribution but dependence under conditioning on K [2]. This conditional induction of dependence is the defining statistical signature of a Collider and is central to explaining selection bias and the “explain-away” effect in probabilistic reasoning.

An **Independent** variable has no causal edges connecting it to X or Y , satisfying $X \nrightarrow K$, $K \nrightarrow X$, $Y \nrightarrow K$, and $K \nrightarrow Y$. Under the Markov assumption, K is statistically independent of both X and Y in the observational distribution. This independence is the strongest of all eight signals since no conditioning manipulation on X or Y can change the marginal distribution of K , yet this very strength makes the Independent class relatively easier to classify correctly, with the first-place solution achieving 86.18% Balanced Accuracy on this class [5].

A **Cause of X** is a variable K that causes X but has no causal relationship with Y , satisfying $K \rightarrow X$ and $K \nrightarrow Y$, $X \nrightarrow K$. This structure is distinguished from a Confounder by the absence of the edge $K \rightarrow Y$; unlike a Confounder, a Cause of X does not induce correlation between X and Y and does not confound the X – Y relationship. Both Cause of X and Confounder share the edge $K \rightarrow X$, making them statistically similar in their K – X marginal relationship, yet they differ in whether K also affects Y .

A **Cause of Y** is a variable K that affects Y directly without being connected to X , satisfying $K \rightarrow Y$ with $K \nrightarrow X$ and $Y \nrightarrow K$. This is the mirror structure of Cause of X with respect to the $X \rightarrow Y$ edge. Cause of Y and Confounder both have $K \rightarrow Y$, but Confounder additionally has $K \rightarrow X$, which distinguishes the two roles and explains why scalar statistics based on pairwise correlations between K and Y alone cannot separate these classes.

A **Consequence of X** satisfies $X \rightarrow K$ with $K \nrightarrow X$, $K \nrightarrow Y$, and $Y \nrightarrow K$. This is the effect structure corresponding to Cause of X in the reverse direction: if K is a Cause of X , then K is a Consequence of Y with respect to the pair (X, Y) , and vice versa. Consequence of X and Mediator both have the edge $X \rightarrow K$, yet they

differ in whether K subsequently affects Y , which is why the Mediator is strictly more challenging to classify: detecting the presence or absence of the edge $K \rightarrow Y$ requires examining the downstream causal effect of K on Y .

A **Consequence of Y** satisfies $Y \rightarrow K$ with the remaining three edges absent. This is the effect structure corresponding to Cause of Y in reverse. Consequence of Y and Independent both result in K being independent of the other variable not in the causal pathway, yet Independent has no edges at all while Consequence of Y has $Y \rightarrow K$, creating detectable dependence between Y and K in the observational distribution.

2.1.3 Challenges in Classifying Causal Roles

The difficulty of classifying the eight roles stems from the fact that several pairs share nearly identical statistical signatures that scalar measures cannot distinguish. Olivetti et al. (2025) analyzed the post-competition results and found that the first-place solution achieved only 65.67% Balanced Accuracy on the Mediator class, the lowest among all eight classes, while reaching 86.18% on Independent [5]. The organizers’ supervised baseline achieved only 13.89% on Mediator and 18.60% on Confounder, confirming these as the two most challenging classes in the competition.

The Confounder versus Cause of X problem exemplifies the challenge of multi-edge discrimination. Both roles have the edge $K \rightarrow X$, which induces correlation between K and X through the same mechanism. The distinguishing signal is whether K also has an edge to Y , yet this requires examining the full causal structure rather than a pairwise relationship. Scalar statistics that measure correlation or dependence between K and X alone cannot detect the presence or absence of $K \rightarrow Y$, which is why models that process each edge independently without graph-level aggregation miss this distinction.

The Mediator versus Consequence of X challenge is equally subtle. Both roles have the edge $X \rightarrow K$, creating dependence between X and K through the same mechanism. The distinguishing feature of a Mediator is that K subsequently affects Y , transmitting part of the causal effect from X to Y . Detecting this requires modeling the functional form of the relationship chain $X \rightarrow K \rightarrow Y$ rather than

merely testing for independence between X and K , which is positive for both roles. The Mediator versus Confounder challenge is particularly instructive because both roles satisfy the conditional independence $X \perp Y \mid K$ through different structural mechanisms [9]. In a Mediator, $X \perp Y \mid K$ holds because conditioning on K blocks the pathway $X \rightarrow K \rightarrow Y$. In a Confounder, the same conditional independence holds because X and Y are conditionally independent given their common cause K via the d-separation property. Scalar independence tests applied after conditioning cannot distinguish these two mechanisms, which explains why even the first-place solution struggles on both classes and why the edge context aggregation described in Chapter 3 is essential for separating them.

2.2 Edge Tensor Representation

2.2.1 Motivation for Edge-Based Processing

The first-place solution introduced a fundamental representational shift: rather than processing the raw data matrix ($n \times p$) with its arbitrary variable ordering, each ordered variable pair (u, v) is represented as a multi-channel edge tensor of length n processed by a neural network [6]. This approach provides two key advantages. First, the architecture becomes permutation-equivariant with respect to variable ordering: permuting the columns of the data matrix permutes the set of edge tensors correspondingly, so the model produces the same output regardless of how variables are labeled. Second, directional information is preserved separately for $(u \rightarrow v)$ and $(v \rightarrow u)$, allowing the model to learn causal asymmetry from the directionality encoded in the two separate tensors.

This edge-centric representation maps naturally onto the causal graph structure, where each variable K participates in four edges: $K \rightarrow X$, $K \rightarrow Y$, $X \rightarrow K$, and $Y \rightarrow K$. Each edge carries different causal information about K 's role, and the four directional embeddings together determine the eight-class prediction. The edge tensor thus serves as the fundamental unit of representation in the proposed architecture.

2.2.2 The Eight Channels of the Edge Tensor

Given an ordered variable pair (u, v) and a permutation σ that sorts u in ascending order, the 8-channel edge tensor is constructed as follows. Channel 1 records the sorted values of u : $\mathbf{c}_1 = (u_{\sigma(1)}, \dots, u_{\sigma(n)})$, preserving the marginal distribution of u . Channel 2 records v reordered by the rank of u : $\mathbf{c}_2 = (v_{\sigma(1)}, \dots, v_{\sigma(n)})$, encoding the monotonic relationship and rank correlation between u and v .

Channels 3 through 5 compute the Nadaraya–Watson kernel regression estimator [10, 11] at each sorted point $u_{\sigma(i)}$ with three bandwidths $h \in \{0.2, 0.5, 1.0\}$:

$$\hat{m}_h(u_{\sigma(i)}) = \frac{\sum_{j=1}^n \exp\left(-\frac{(u_{\sigma(i)} - u_j)^2}{2h^2}\right) v_j}{\sum_{j=1}^n \exp\left(-\frac{(u_{\sigma(i)} - u_j)^2}{2h^2}\right)}. \quad (2.1)$$

Each bandwidth captures a different smoothness scale of the regression function. The smallest bandwidth $h = 0.2$ is sensitive to local structure and captures fine-grained nonlinearities; the medium bandwidth $h = 0.5$ balances local and global patterns; the largest bandwidth $h = 1.0$ reflects global trends and is robust to outliers. Multi-bandwidth representation prevents the model from depending on any single choice of h , which is critical because the optimal smoothness varies across different causal mechanisms [11].

Channels 6 through 8 compute ANM residuals with the same three bandwidths: $r_h(u_{\sigma(i)}) = v_{\sigma(i)} - \hat{m}_h(u_{\sigma(i)})$. The Additive Noise Model assumes $V = f(U) + \varepsilon$ with $\varepsilon \perp U$ and posits that the true causal direction $U \rightarrow V$ implies $V - \hat{m}(U) \perp U$, whereas the reverse direction does not satisfy this independence property [12, 13]. The causal asymmetry of ANM residuals is what makes these channels informative: under the correct causal direction, residuals are approximately independent of the cause, producing a characteristic flat distributional signature that a Conv1D network can learn to detect. Under the wrong direction, residuals exhibit systematic structure that betrays the incorrect assignment.

2.2.3 From 1D Curves to 2D Density Images

The 1D edge tensor representation encodes directional relationships between variable pairs as functional curves, capturing the statistical signature of potential causal edges. However, this representation is inherently pairwise: it describes the marginal

relationship between u and v separately for each ordered pair rather than capturing the joint visual pattern of a variable K with respect to both X and Y simultaneously. This limitation motivates the introduction of a two-dimensional scatter density representation that complements the 1D edge tensors.

Given n observation pairs (a_i, b_i) for two variables a and b , a two-dimensional histogram partitions the joint domain $[\min_a, \max_a] \times [\min_b, \max_b]$ into a $G \times G$ grid and counts observations in each bin [14]:

$$H[r, c] = \#\{i : a_i \in \text{bin}_r, b_i \in \text{bin}_c\}, \quad r, c \in \{1, \dots, G\}. \quad (2.2)$$

This matrix $H \in \mathbb{Z}^{G \times G}$ is a discrete approximation to the joint probability mass function $p(a, b)$. Raw histograms are noisy at finite sample size; to produce a continuous, visually interpretable density image, each histogram is convolved with a Gaussian kernel of standard deviation σ (in bin units) [15]:

$$\hat{H}_\sigma[r, c] = \sum_{r', c'} H[r', c'] \cdot \frac{1}{2\pi\sigma^2} \exp\left(-\frac{(r - r')^2 + (c - c')^2}{2\sigma^2}\right). \quad (2.3)$$

The smoothed image $\hat{H}_\sigma \in \mathbb{R}^{G \times G}$ represents the estimated joint density $\hat{p}(a, b)$. A central insight motivating this representation is that conditional independence has a visual signature in scatter density images: under the ANM assumption, if the true causal direction is $u \rightarrow v$, then the residual $r_{u \rightarrow v} = v - \hat{m}(u)$ is approximately independent of u , and the scatter plot of $(u, r_{u \rightarrow v})$ appears as a uniform horizontal band with no visible structure [13]. Conversely, the residual scatter under the wrong causal direction exhibits systematic curvature or heteroscedasticity. The 2D convolutional network learns to detect this directional asymmetry in the density image.

2.3 Deep Learning Components

2.3.1 Convolutional and Residual Networks

One-dimensional convolutional networks apply learned filters over 1D sequences, making them well-suited to extract local features from edge tensors where each edge is a sequence of length $n = 1000$ [16]. With a filter of width k , a Conv1D layer computes $y_t = \sum_{j=0}^{k-1} w_j \cdot x_{t+j} + b$, producing a feature representation at each

position in the sequence. Two-dimensional convolution extends this to spatial data with two axes, such as scatter density images, applying a learned filter $\mathbf{W} \in \mathbb{R}^{k \times k}$ to an input feature map $\mathbf{X}$ to produce spatial feature representations that preserve local connectivity and translation equivariance.

Deep networks face the vanishing gradient problem as depth increases, which residual connections address by adding a skip connection: $\mathbf{y} = \mathcal{F}(\mathbf{x}; \{W_i\}) + \mathbf{x}$, where $\mathcal{F}$ is the learned mapping and $\mathbf{x}$ is the shortcut input [16]. This identity mapping ensures that the gradient flows directly through the network, enabling successful training of very deep architectures. Each residual block combines Group Normalization [17] (chosen over Batch Normalization for stability with small batch sizes), the GELU activation function [18] which approximates $\text{GELU}(x) = x \cdot \Phi(x)$ with the standard normal cumulative distribution function Φ , and a Conv1D or Conv2D layer with the residual shortcut.

2.3.2 Self-Attention and Multi-Head Attention

After each edge tensor is processed through Conv1D blocks and reduced to a representation vector $\mathbf{e}_{uv} \in \mathbb{R}^d$ via Global Average Pooling, information from all edges in the graph must be aggregated to reason about each variable’s role. The self-attention mechanism allows each element in a sequence to attend to all other elements [19]:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V, \quad (2.4)$$

where $Q = \mathbf{X}W^Q$, $K = \mathbf{X}W^K$, $V = \mathbf{X}W^V$ are linear projections. Multi-head attention runs H attention heads in parallel:

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_H) W^O, \quad (2.5)$$

allowing each head to attend to different representation subspaces.

Graph Attention Networks extend the attention mechanism to graph-structured data by computing attention weights over edges, allowing each node to attend to its neighbors with edge-specific weights learned dynamically from features [20]. Relational inductive biases refer to architectural constraints that encode assumptions about how entities in a system relate to each other; in causal graphs, specific topological patterns (chains, forks, colliders) carry semantic meaning that structured attention biases can encode directly [21].

2.3.3 Loss Functions and Training Strategy

The model employs a dual-head architecture with two loss functions. The Node head performs 8-class classification and uses Cross-Entropy loss with inverse-frequency class weights to handle class imbalance. The Edge head is a binary classification (causal edge present or not) used as an auxiliary loss during training only. The combined loss is $\mathcal{L} = \mathcal{L}_{\text{node}} + \lambda \cdot \mathcal{L}_{\text{edge}}$, where λ is a balancing hyperparameter [22]. Optimization uses AdamW [23] with cosine annealing scheduling [24]. The Focal Loss variant down-weights easy examples and focuses training on hard misclassified cases, which is particularly beneficial for the Mediator and Confounder classes where the competition analysis showed models struggle most [5].

2.4 Existing Approaches on the ADIA Lab Challenge

2.4.1 First-Place Solution: End-to-End Deep Learning

The first-place solution introduced an end-to-end deep learning model with the following architecture: 3-channel edge tensor construction $\rightarrow$ Stem Conv1D (3 $\rightarrow$ 64 channels) $\rightarrow$ 5 Residual Conv1D blocks $\rightarrow$ Global Average Pooling $\rightarrow$ Merge with edge-type embedding $\rightarrow$ 2 Self-Attention layers $\rightarrow$ Node head (8-class) plus Edge head (binary, auxiliary) [6]. The model has approximately 188K parameters and achieves 76.70% Balanced Accuracy on the public leaderboard, establishing a strong baseline that outperforms all classical and ML feature engineering approaches. Olivetti et al. (2025) provide a comprehensive post-competition analysis confirming that the deep learning approach substantially outperforms conditional independence tests and gradient boosting on the fine-grained eight-class problem [5].

2.4.2 Rank 4–5 Solutions: Feature Engineering with Gradient Boosting

The fourth-place and fifth-place solutions follow a feature engineering and gradient boosting approach [25, 26]. These solutions extract statistical features including Pearson and Spearman correlation coefficients, partial correlations, conditional mutual information, and HSIC scores, together with the outputs of classical causal discovery algorithms such as LiNGAM [12], NOTEARS, PC, and GES. An XGBoost or LightGBM ensemble then performs the 8-class classification. The advantages

of this approach are interpretability and no GPU requirement; the disadvantages are dependence on manual feature engineering and the inability to learn end-to-end representations from raw data. Despite extracting over 300 features, the ML pipeline achieves only 72.64% Balanced Accuracy, trailing the deep learning baseline by 4.32 percentage points.

2.4.3 Position of This Thesis

This thesis reimplements the first-place solution in detail, achieving 73.96% local and 73.61% public leaderboard Balanced Accuracy on the faithful baseline reimplementation, confirming reproducibility of the deep learning approach. The specific technical contributions extend this baseline through five targeted improvements that address the performance gaps on difficult classes, raising Balanced Accuracy to 81.082%. The contributions are presented in detail in Chapter 3.

2.5 Data Augmentation for Causal Discovery

2.5.1 Label-Preserving Transformations

Data augmentation increases effective training data without additional collection cost by applying label-preserving transformations to existing samples. In image classification, standard augmentations include random cropping, rotation, and color jittering because the object label remains unchanged under these geometric or photometric transformations. For graph-structured data, augmentation exploits symmetries in the causal representation. A causal graph remains semantically equivalent under certain transformations: relabeling variables, permuting node orderings, or treating different edges as the treatment-outcome pair of interest.

Multi-view learning provides theoretical grounding for this approach: when multiple views of the same underlying structure are available, training on all views can improve generalization by exposing the model to different perspectives of the same causal relationships [27]. In the context of the ADIA Lab Challenge, each choice of the treatment-outcome pair (X, Y) provides a different view of the same DAG structure, with the eight-class labels recomputed relative to the chosen pair.

2.5.2 X/Y Remap as Data Multiplication

The symmetry of a causal DAG implies that any directed edge $A \rightarrow B$ could serve as the designated treatment-outcome pair (X, Y) , with the remaining variables classified relative to this choice. The X/Y remap augmentation treats each directed edge in the ground-truth DAG as a new training sample by selecting it as the treatment-outcome pair, recomputing all eight-class labels for every other variable in the graph. This transformation is label-preserving in the sense that the ground-truth causal structure is unchanged; only the framing of which pair serves as (X, Y) changes.

For a graph with e directed edges, this produces approximately e new labeled samples from each original sample, where e varies with the number of variables p and the edge density of the DAG. On average across the training set, this yields approximately an 11-fold increase in effective training data, with the largest gains for graphs that have more edges (more variables and denser causal structure). The augmentation is particularly beneficial for rare classes such as Mediator and Collider, where the additional samples from edges that involve these roles help address the class imbalance problem [5].

Chapter 3

Neural Edge-Node Fusion: A Dual-Pipeline Architecture for Causal Discovery

This chapter describes the complete pipeline of the model proposed in this thesis. The method takes as input an observational dataset of $n = 1000$ observations over $p \in \{3, \dots, 10\}$ variables – with a designated treatment variable X and outcome variable Y – and produces an 8-class causal role label for each remaining variable $K \notin \{X, Y\}$.

The proposed model builds on the edge-centric deep learning framework introduced by the first-place solution of the ADIA Lab Causal Discovery Challenge [5, 6], and extends it with five targeted contributions: the first contribution expands the edge tensor from 3 to 8 channels through multi-bandwidth kernel regression and ANM residuals, capturing causal asymmetry at multiple scales with a gain of +1.89%; the second contribution injects topology-aware priors into self-attention layers using only six learned scalars, encoding chain, fork, and collider relationships directly to achieve +2.65%; the third contribution introduces a 2D scatter density pipeline that renders joint distributions as visual images, capturing patterns invisible to pairwise analysis with +0.88%; the fourth contribution uses a $\lambda = 0.7$ dual-loss weighting scheme that ensures edge embeddings remain causally meaningful throughout training, adding +0.59%; and the fifth contribution applies X/Y remap data augmentation that multiplies the training set by approximately 11x without additional data collection, contributing +1.55% to +3.32% depending on configuration.

3.1 Architecture Overview

The model consists of two parallel processing pipelines that operate on different representations of the same observational data, followed by a fusion step for final classification. Figure 3.1 provides the complete architectural diagram illustrating how these components connect.

The **Edge Context Encoder** (Section 3.3) processes all $p(p-1)$ directed variable pairs simultaneously. Each ordered pair (u, v) is represented as an 8-channel 1D tensor of length n , capturing the statistical and causal relationship between u and v through sorted observations, multi-bandwidth kernel regression curves, and ANM residuals. These tensors are processed independently by a shared-weight Conv1D network, then aggregated across the full graph via structural self-attention. A binary edge head supervises this pipeline during training, forcing the edge embeddings to encode causally meaningful graph structure.

The **Node Visual Classifier** (Section 3.5) processes each non- X/Y variable K individually. For each such node, an 8-channel 32×32 scatter density image is constructed by rendering the joint distribution of K against X and Y (and their ANM residuals) as smoothed 2D histograms. A hierarchical Conv2D network extracts visual features from this image.

The **Fusion** step (Section 3.6) combines the node’s 2D visual embedding with the four edge context embeddings corresponding to edges $K \rightarrow X$, $K \rightarrow Y$, $X \rightarrow K$, and $Y \rightarrow K$, producing the final 8-class prediction for each node.

3.2 Preprocessing and Edge Tensor Construction

This section describes the construction of the 8-channel edge tensor representation, which is the first contribution of this thesis extending the 3-channel baseline.

3.2.1 Data Normalization

Each input dataset is a matrix $\mathbf{D} \in \mathbb{R}^{n \times p}$ with $n = 1000$ observations. Prior to feature construction, each column (variable) is standardized to zero mean and unit variance via z-score normalization:

$$\tilde{x}_i = \frac{x_i - \mu}{\sigma + \varepsilon}, \quad (3.1)$$

where μ and σ are the column mean and standard deviation respectively, and $\varepsilon = 10^{-8}$ prevents division by zero. Normalization ensures that all subsequent computations – particularly kernel bandwidth selection – are scale-invariant across variables and datasets.

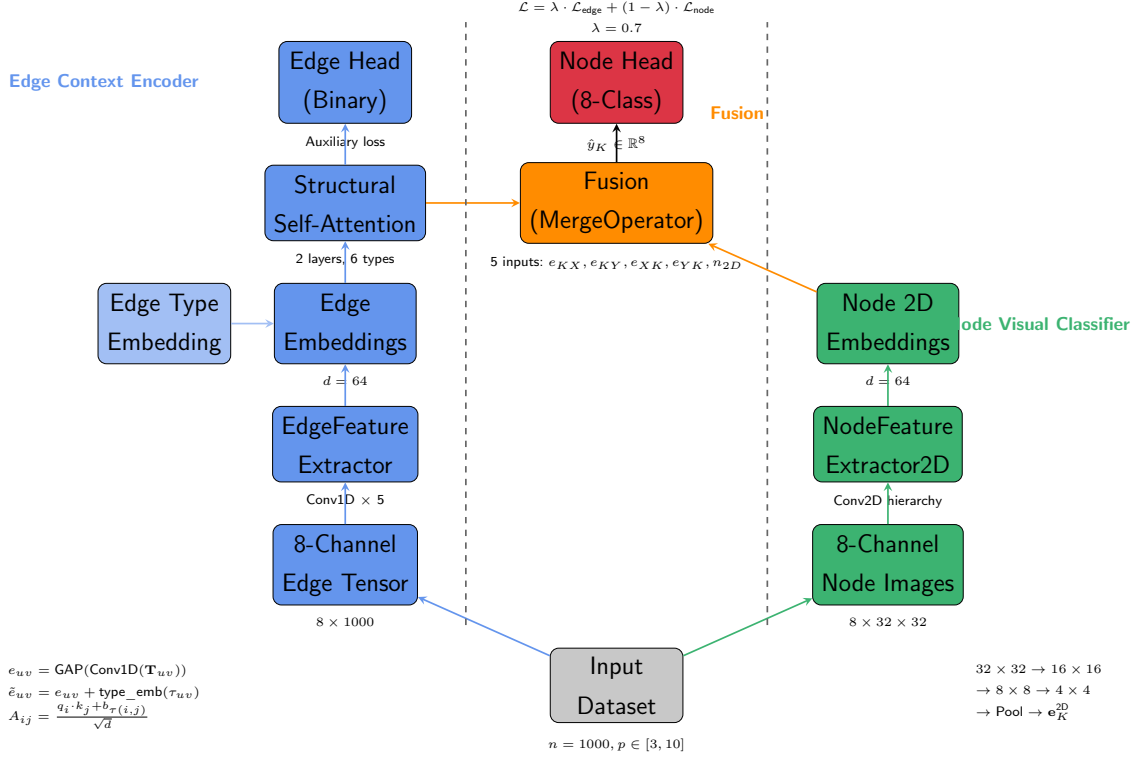


Figure 3.1: The v26e architecture: dual-pipeline neural network combining 1D edge context encoding with 2D scatter density visual features. The Edge Context Encoder (left, blue) processes all $p(p - 1)$ directed edges through Conv1D and structural self-attention. The Node Visual Classifier (right, green) extracts features from 8-channel 32×32 density images. The Fusion layer (center, orange) combines four edge embeddings ($K \rightarrow X, K \rightarrow Y, X \rightarrow K, Y \rightarrow K$) with the 2D visual embedding for final 8-class classification.

3.2.2 Complete Directed Graph and Edge Pairs

From p variables, a complete directed graph with $p(p - 1)$ ordered pairs is constructed. Every ordered pair (u, v) with $u \neq v$ produces a separate edge tensor: the pairs (u, v) and (v, u) are treated as distinct, because the directional asymmetry between them is a primary causal signal. This permutation-equivariant design ensures that classification results are independent of the arbitrary naming order of variables.

3.2.3 8-Channel Edge Tensor

For each ordered pair (u, v) , let σ denote the permutation that sorts variable u in ascending order, so that $u_{\sigma(1)} \leq \dots \leq u_{\sigma(n)}$. The 8-channel edge tensor $\mathbf{T}_{uv} \in \mathbb{R}^{8 \times n}$ is constructed as follows. Figure 3.2 illustrates the resulting curves for each channel.

Channels 1–2: Sorted observations. Channel 1 preserves the marginal distribution of u : $\mathbf{c}_1 = (u_{\sigma(1)}, \dots, u_{\sigma(n)})$. Channel 2 records v reordered by the rank of u : $\mathbf{c}_2 = (v_{\sigma(1)}, \dots, v_{\sigma(n)})$, encoding the functional relationship between u and v as an empirical curve. When sorted by rank of u , channel 2 traces the conditional expectation $\mathbb{E}[v \mid u]$: monotone relationships appear as smooth increasing or decreasing curves, non-monotone relationships as curved or U-shaped traces. This representation retains the full shape of the conditional relationship, which is information irreversibly lost by scalar summaries such as correlation coefficients.

Channels 3–5: Multi-bandwidth kernel regression. At each sorted point $u_{\sigma(i)}$, the Nadaraya–Watson estimator [10, 11] computes:

$$\hat{m}_h(u_{\sigma(i)}) = \frac{\sum_{j=1}^n \exp\left(-\frac{(u_{\sigma(i)} - u_j)^2}{2h^2}\right) v_j}{\sum_{j=1}^n \exp\left(-\frac{(u_{\sigma(i)} - u_j)^2}{2h^2}\right)}, \quad (3.2)$$

with three bandwidths $h \in \{0.2, 0.5, 1.0\}$, yielding channels $\mathbf{c}_3, \mathbf{c}_4, \mathbf{c}_5$. Each bandwidth captures a different smoothness scale of the regression function: $h = 0.2$ is sensitive to local curvature and sharp transitions, while $h = 1.0$ reflects global monotone trends. Using all three bandwidths together prevents the model from depending on a single fixed scale assumption, which is critical because the optimal smoothness varies across different causal mechanisms.

Channels 6–8: Multi-bandwidth ANM residuals. The Additive Noise Model [13]

posits that if $U \rightarrow V$ is the true causal direction, the residual $V - \hat{m}(U)$ should be approximately independent of U ; the reverse-direction residual retains structure. This asymmetry is encoded in channels 6–8:

$$r_h(u_{\sigma(i)}) = v_{\sigma(i)} - \hat{m}_h(u_{\sigma(i)}), \quad h \in \{0.2, 0.5, 1.0\}, \quad (3.3)$$

yielding channels $\mathbf{c}_6, \mathbf{c}_7, \mathbf{c}_8$. When $u \rightarrow v$ holds, the residual curve is approximately flat (independent of u); under the reversed direction, the residual retains a systematic nonlinear shape. The Conv1D network learns to detect this difference directly from the curve shape.

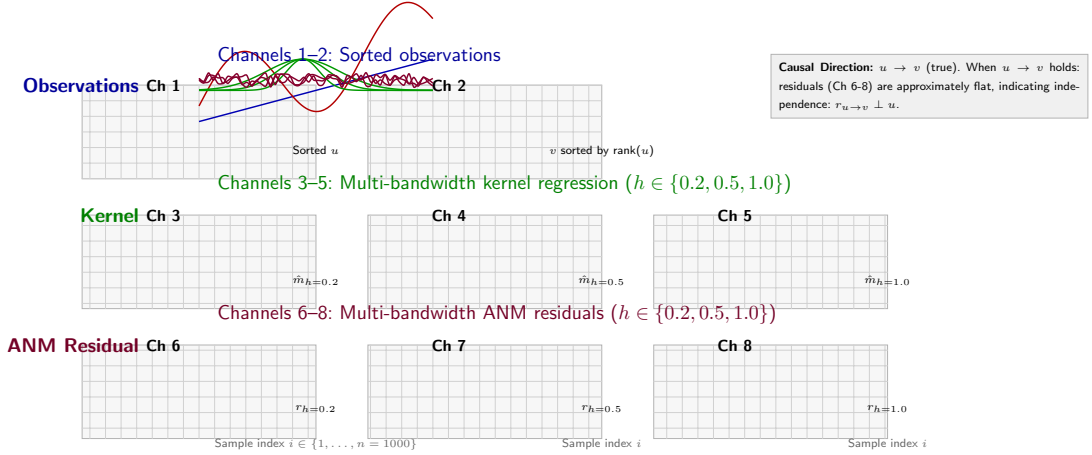


Figure 3.2: The 8-channel edge tensor $\mathbf{T}_{uv} \in \mathbb{R}^{8 \times n}$ for a variable pair (u, v) . **Row 1** (blue): Sorted observations – Channel 1 preserves the marginal distribution of u , Channel 2 shows v sorted by rank of u revealing functional relationship. **Row 2** (green): Nadaraya-Watson kernel regression at three bandwidths capturing different smoothness scales ($h = 0.2$ local, $h = 1.0$ global). **Row 3** (purple): ANM residuals – approximately flat when $u \rightarrow v$ is the true direction (independent noise), structured in reverse direction.

3.3 Edge Context Encoder

This section describes the 1D pipeline that processes all edge tensors and aggregates them into node-level representations through structural self-attention.

3.3.1 Per-Edge Feature Extraction via Conv1D

The 8-channel edge tensor $\mathbf{T}_{uv} \in \mathbb{R}^{8 \times n}$ is processed by a shared-weight Conv1D network to produce a $d = 64$ dimensional edge embedding. The network consists of

a stem layer followed by five residual blocks. The stem layer expands the 8 input channels to $d = 64$ hidden channels:

$$\mathbf{H}_0 = \text{Conv1D}_{8 \rightarrow 64}(\mathbf{T}_{uv}) \in \mathbb{R}^{64 \times n}. \quad (3.4)$$

Each of the five residual Conv1D blocks applies the pattern from He et al.[16], adapted with Group Normalization [17] for stability under small batch sizes and the GELU activation [18]:

$$\mathbf{H}_{l+1} = \mathbf{H}_l + \text{Conv1D}\left(\text{GELU}\left(\text{GroupNorm}(\mathbf{H}_l)\right)\right). \quad (3.5)$$

After the five residual blocks, Global Average Pooling reduces the sequence dimension, producing a fixed-size representation regardless of n :

$$\mathbf{e}_{uv}^{\text{conv}} = \text{GAP}(\mathbf{H}_5) = \frac{1}{n} \sum_{t=1}^n \mathbf{H}_5[:, t] \in \mathbb{R}^{64}. \quad (3.6)$$

3.3.2 Edge Type Embedding and Merge

Each directed edge (u, v) is assigned a discrete edge type τ_{uv} from a 7-class taxonomy based on the positions of u and v relative to X and Y . A learnable embedding table maps each type to a vector $\mathbf{t}_{\tau_{uv}} \in \mathbb{R}^{64}$. The final edge representation combines the Conv1D embedding with the type embedding via addition, which preserves the representation dimension:

$$\tilde{\mathbf{e}}_{uv} = \mathbf{e}_{uv}^{\text{conv}} + \mathbf{t}_{\tau_{uv}}. \quad (3.7)$$

3.3.3 Binary Edge Head

A linear layer projects each $\tilde{\mathbf{e}}_{uv}$ to 2 logits corresponding to the binary label of whether the directed edge $u \rightarrow v$ exists in the ground-truth DAG. This edge head is used only during training as an auxiliary supervision signal; its predictions are discarded at inference. The role of this auxiliary supervision is discussed in Section 3.7.

3.4 Structural Attention Bias

This section describes the second contribution of this thesis: a topology-aware attention mechanism that injects causal structural priors into self-attention with only six learnable scalars per head.

3.4.1 Motivation

Standard multi-head self-attention assigns attention weights between all pairs of edge representations purely from learned query-key dot products, with no consideration of their topological relationship in the causal graph. This ignores a strong prior: edges that share an endpoint or form a directed chain through a node of interest are structurally more relevant to each other than unrelated background edges. For example, in the chain $X \rightarrow K \rightarrow Y$, the edges $X \rightarrow K$ and $K \rightarrow Y$ should attend to each other more strongly than to a background edge $A \rightarrow B$.

3.4.2 Six Topological Relationship Types

For each ordered pair of edges (i, j) in the graph, a topological relationship type $\tau(i, j) \in \{0, \dots, 5\}$ is assigned based on shared endpoints. Given edges $e_i = (u_1, v_1)$ and $e_j = (u_2, v_2)$, the six types are defined in Table 3.1. Each type carries a distinct causal interpretation: reverse pairs represent the same variable pair in opposite directions (causal asymmetry signal); shared source pairs indicate a common cause structure (fork); shared target pairs indicate a common effect structure (collider); forward and backward chain pairs indicate mediator pathways; unrelated pairs carry no topological structure.

Table 3.1: Six topological relationship types between edge pairs in the structural attention bias. Each type is determined by whether the two edges share a source or target node, encoding a distinct causal structural relationship. The learned scalar bias b_τ allows the model to selectively strengthen or weaken attention between edge pairs according to their causal topology.

Type	Condition	Causal interpretation
Reverse	$u_1 = v_2, v_1 = u_2$	Same pair, opposite direction
Shared source	$u_1 = u_2, v_1 \neq v_2$	Common cause (fork) structure
Shared target	$v_1 = v_2, u_1 \neq u_2$	Common effect (collider) structure
Forward chain	$v_1 = u_2$	Mediator chain $u_1 \rightarrow v_1 \rightarrow v_2$
Backward chain	$v_2 = u_1$	Reverse mediator chain
Unrelated	otherwise	No shared endpoint

3.4.3 Learned Scalar Bias per Topological Type

The attention logit between edges i and j is augmented with a learnable scalar bias specific to their topological relationship:

$$A_{ij} = \frac{\mathbf{q}_i \cdot \mathbf{k}_j + b_{\tau(i,j)}}{\sqrt{d}}, \quad (3.8)$$

where $\mathbf{q}_i, \mathbf{k}_j \in \mathbb{R}^d$ are the standard query and key projections, d is the head dimension, and $b_{\tau} \in \mathbb{R}$ is a single learnable scalar for each of the six types. This introduces only 6 additional parameters per attention head while encoding explicit causal structural prior knowledge. All bias scalars are initialized to zero, so training begins from standard unbiased self-attention, preserving learned representations from early epochs while gradually adjusting attention weights according to causal topology. Two layers of multi-head self-attention with structural bias are applied sequentially to aggregate information across the full edge set.

3.4.4 Why Structured Bias Outperforms Unconstrained Capacity

Several other architectural extensions – parallel observation cross-attention branches, observation-level transformers, node-centric aggregation – all produce results at or below the 3-channel baseline. The distinguishing factor is that structural attention bias adds a structured inductive prior with a minimal parameter budget (6 scalars per head), rather than additional unconstrained capacity. The bias does not learn a new representation from scratch – it only learns the relative priority among topologically distinct edge pairs. This is validated by the experimental result: structural bias improves Balanced Accuracy by +2.65% over the same input channels without it, a gain larger than any other single architectural change in this thesis.

3.5 Node Visual Representation

This section describes the third contribution of this thesis: a 2D scatter density pipeline that provides complementary visual features to the 1D edge tensor representation.

3.5.1 Motivation

The 1D edge tensor representation is inherently pairwise: it describes the marginal relationship between u and v rather than the joint visual pattern of a node K with respect to both X and Y simultaneously. The node visual representation complements the 1D pipeline by rendering the joint scatter distribution of K against X and Y as a two-dimensional density image. This captures visual spatial structure – clustering, curvature, heteroscedasticity, and residual patterns – that is invisible in any single 1D marginal curve.

3.5.2 8-Channel Scatter Density Image

For each non- X/Y node K , an 8-channel 32×32 image $\mathbf{I}_K \in \mathbb{R}^{8 \times 32 \times 32}$ is constructed from the $n = 1000$ observations. The 8 channels are grouped into two sets with separate smoothing parameters.

Channels 1–4 render the raw scatter density with $\sigma_{\text{raw}} = 4.0$: $\mathbf{I}_K^{(1)} = \rho(K, X)$, $\mathbf{I}_K^{(2)} = \rho(K, Y)$, $\mathbf{I}_K^{(3)} = \rho(X, K)$, $\mathbf{I}_K^{(4)} = \rho(Y, K)$, where $\rho(a, b)$ denotes the smoothed 32×32 joint density image of variables a (horizontal) and b (vertical), each axis min-max normalized to $[-1, 1]$. The four views provide the full bidirectional visual context of K with respect to X and Y . A larger smoothing $\sigma_{\text{raw}} = 4.0$ is used to smooth out observation noise while preserving the coarse spatial structure of the joint distribution.

Channels 5–8 encode directional asymmetry with $\sigma_{\text{ANM}} = 2.0$: $\mathbf{I}_K^{(5)} = \rho(K, r_{X \rightarrow K})$, $\mathbf{I}_K^{(6)} = \rho(X, r_{K \rightarrow X})$, $\mathbf{I}_K^{(7)} = \rho(K, r_{Y \rightarrow K})$, $\mathbf{I}_K^{(8)} = \rho(Y, r_{K \rightarrow Y})$, where $r_{u \rightarrow v}$ denotes the ANM residual of v regressed on u . If the true causal direction is $u \rightarrow v$, the residual $r_{u \rightarrow v}$ is approximately independent of u , producing a uniform horizontal scatter; the reverse-direction residual retains visible nonlinear structure. A finer smoothing $\sigma_{\text{ANM}} = 2.0$ is used to preserve the fine-grained directional asymmetry patterns that coarser smoothing would obscure.

An earlier variant with 12 channels (adding kernel coefficient maps) scored 80.42% versus 80.47% for the 8-channel version. The kernel coefficient represents a derived linear relationship with no meaningful spatial structure along the observation axis, making Conv2D’s spatial inductive bias inappropriate for these channels.

3.5.3 Hierarchical Conv2D Feature Extractor

The 8-channel 32×32 image $\mathbf{I}_K$ is processed by a hierarchical Conv2D network that progressively downsamples the spatial resolution: $32 \times 32 \rightarrow 16 \times 16 \rightarrow 8 \times 8 \rightarrow 4 \times 4 \rightarrow \mathbf{e}_K^{2D} \in \mathbb{R}^{64}$. Each Conv2D block uses Group Normalization and GELU activations consistent with the 1D pipeline. The final AdaptiveAvgPool collapses the 4×4 spatial map to a $d = 64$ dimensional node visual embedding.

3.6 Fusion and Node Classification

This section describes how the edge context and visual representations are combined for the final 8-class prediction.

3.6.1 Combining Edge Context with Visual Representation

For each node K , the final classification embedding is formed by combining five sources of information: the four directional edge context embeddings from the Edge Context Encoder and the visual embedding from the Node Visual Classifier. Let $\mathbf{e}_{K \rightarrow X}, \mathbf{e}_{K \rightarrow Y}, \mathbf{e}_{X \rightarrow K}, \mathbf{e}_{Y \rightarrow K} \in \mathbb{R}^{64}$ be the four attended edge embeddings, and let $\mathbf{e}_K^{2D} \in \mathbb{R}^{64}$ be the node’s visual embedding. These are combined via a learned linear merge operator:

$$\mathbf{h}_K = \text{MergeOperator}(\mathbf{e}_{K \rightarrow X}, \mathbf{e}_{K \rightarrow Y}, \mathbf{e}_{X \rightarrow K}, \mathbf{e}_{Y \rightarrow K}, \mathbf{e}_K^{2D}) \in \mathbb{R}^{64}. \quad (3.9)$$

The node head then maps $\mathbf{h}_K$ to 8-class logits:

$$\hat{\mathbf{y}}_K = \text{Linear}_{64 \rightarrow 8}(\mathbf{h}_K). \quad (3.10)$$

3.6.2 Necessity of Edge Context

Table 5.2 presents ablation results confirming that the edge context is structurally necessary, not merely supplementary. When the entire 1D edge pipeline is removed and nodes are classified from their 2D visual embeddings alone, Balanced Accuracy collapses to 51.51% – a drop of nearly 29 percentage points from the full model’s 80.47%. A variant using mean pooling of all node embeddings as a substitute for edge context (v28b) achieves only 46.39%. This confirms that the 2D visual

representation captures local distributional structure of K relative to X and Y , but cannot reason about K 's causal role within the larger graph – a task that requires the global relational information encoded by the attended edge embeddings.

Table 3.2: Ablation study on the necessity of edge context for node classification. The full model (v26e) combines 1D edge context with 2D visual features, achieving 80.47% Balanced Accuracy. Removing the 1D edge pipeline and using 2D visual features alone (v26c) collapses performance to 51.51%, a 28.96-point drop. Replacing edge context with mean-pooled node embeddings (v28b) further degrades to 46.39%. The binary edge loss, by supervising the edge pipeline with ground-truth adjacency matrix entries, ensures these embeddings remain causally well-structured throughout training.

Variant	Balanced Accuracy	Gap to Full
v26e (full model)	80.47%	–
v28b (2D only + mean pool)	46.39%	–34.08%
v26c (2D only)	51.51%	–28.96%

3.7 Dual-Head Loss Weighting

This section describes the fourth contribution of this thesis: a dual-head loss weighting scheme that ensures the edge context encoder remains causally well-supervised throughout training.

3.7.1 Dual-Head Loss with Lambda Weighting

The model is trained with two loss terms. The primary node loss $\mathcal{L}_{\text{node}}$ is a weighted cross-entropy over the 8 causal role classes, with class weights set inversely proportional to class frequency in the training set:

$$\mathcal{L}_{\text{node}} = - \sum_{c=1}^8 w_c \cdot \mathbf{1}[y = c] \cdot \log \hat{p}_c, \quad w_c \propto \frac{1}{n_c}, \quad (3.11)$$

where n_c is the number of training samples of class c , y is the ground-truth label, and $\hat{p}_c$ is the predicted probability for class c . The auxiliary edge loss $\mathcal{L}_{\text{edge}}$ is a binary weighted cross-entropy over all $p(p-1)$ directed edges, predicting whether

each edge exists in the ground-truth DAG. The total training loss is:

$$\mathcal{L} = \lambda \cdot \mathcal{L}_{\text{edge}} + (1 - \lambda) \cdot \mathcal{L}_{\text{node}}, \quad \lambda = 0.7. \quad (3.12)$$

3.7.2 Rationale for $\lambda = 0.7$

The choice of $\lambda = 0.7$ assigns the majority of the training gradient to the auxiliary edge loss. This is counterintuitive – the edge head is discarded at inference and the primary objective is node classification – but it is empirically necessary. The 2D node classifier receives its structural context entirely from the attended edge embeddings. If λ is small (node loss dominates early training), the edge pipeline receives insufficient gradient and converges to representations optimized for the node head’s immediate needs rather than for causally meaningful edge structure, producing degenerate context that degrades the node merger quality. Increasing λ to 0.7 ensures the edge embeddings are well-supervised throughout training. This mechanism is confirmed by ablation: a configuration favoring node loss ($\lambda = 0.3$) achieves 78.77%, while $\lambda = 0.7$ achieves 80.47% – a difference of +1.70% with identical input channels and architecture, differing only in the loss weighting.

3.8 X/Y Remap Data Augmentation

This section describes the fifth contribution of this thesis: a data augmentation strategy that multiplies the effective training set by approximately 11x without additional data collection.

3.8.1 Strategy

The training set contains 23,500 dataset instances, each with a designated pair (X, Y) satisfying $X \rightarrow Y$ in the ground-truth DAG. However, a dataset instance with p variables contains multiple valid directed edges in the ground-truth DAG, each of which could serve as a treatment-outcome pair. The X/Y remap augmentation exploits this symmetry: for each directed edge $A \rightarrow B$ in the ground-truth DAG, a new training sample is generated by treating (A, B) as the new (X', Y') and recomputing 8-class labels for all remaining nodes K' according to their causal relationship with A and B . This produces approximately 11.2x more training sam-

ples, yielding a training set of approximately 263,000 samples. The augmentation is justified by the symmetry of the classification problem: the 8 causal roles are defined purely by the local structure of the graph around the designated (X, Y) pair, and this local structure occurs with equal validity for any directed edge in the DAG.

3.8.2 Efficient Implementation

Naively recomputing the 8-channel edge tensors for each remap would multiply the preprocessing cost by approximately 11x. The key efficiency insight is that the kernel regression computation (channels 3–8) depends only on the observation matrix $\mathbf{D}$, not on the choice of (X, Y) . Therefore, coefficient maps and residual maps are computed once per graph. For each remap, only three operations are required: edge type re-assignment (cheap name-substitution), node image reconstruction from cached maps (cheap histogram operations), and label re-derivation from the adjacency matrix (a lookup table query). No kernel regression is recomputed.

Gains from X/Y remap augmentation are configuration-dependent. On v8b (3-channel baseline), augmentation provides +3.32% improvement. On v11 (8-channel, structural attention), augmentation provides +1.55%. On v26e (full dual-pipeline), augmentation provides +0.72%. The diminishing returns reflect that richer base representations benefit less from additional data multiplication, though the augmentation remains beneficial across all configurations.

3.9 Training Configuration

All models are trained with the following fixed configuration: AdamW optimizer [23] with cosine annealing learning rate schedule [24], initial learning rate 10^{-3} , batch size 64, 10 training epochs on the augmented 263K dataset (equivalent to approximately 110 epochs over the base 23,500 samples), and 32-bit floating-point precision throughout. A Gaussian noise augmentation of standard deviation 0.01 is applied to edge tensors during training to improve generalization. Training uses Distributed Data Parallel across two NVIDIA RTX 5880 GPUs. The model has approximately 280K parameters in total (edge encoder: approximately 188K; node

Conv2D extractor: approximately 92K; node merge and head: negligible).

Chapter Summary

This chapter presented the five contributions that extend the first-place deep learning baseline from 76.70% to 81.082% Balanced Accuracy. The first contribution, an 8-channel edge tensor, extends the 3-channel representation with multi-bandwidth kernel regression and ANM residuals, preserving the full shape of conditional relationships that scalar features irreversibly discard. The second contribution, a structural attention bias, injects topology-aware causal priors into self-attention with only six learned scalars, encoding chain, fork, and collider relationships directly and outperforming failed experiments that added thousands of unconstrained parameters. The third contribution, a 2D scatter density pipeline, complements the 1D edge tensors by rendering joint distributions as visual images that capture multimodality, heteroscedasticity, and residual patterns invisible to pairwise analysis. The fourth contribution, a dual-head loss weighting scheme with $\lambda = 0.7$, ensures the edge context encoder remains causally well-supervised throughout training, producing richer structural embeddings that the node classifier depends on. The fifth contribution, X/Y remap data augmentation, multiplies the training set by approximately 11x without additional data collection, with the largest gains on difficult minority classes. The combined system achieves 81.082% Balanced Accuracy on the CrunchDAO private test set.

Chapter 4

Evaluation Methodology

This chapter describes the dataset, evaluation metric, experimental baselines, and procedures used to assess the proposed method. The evaluation framework is designed to provide trustworthy performance estimates through local validation before any public leaderboard submission, ensuring that architectural decisions are grounded in evidence rather than overfitting to the public test set.

4.1 Dataset

4.1.1 Data Source and Split

All experiments use the official dataset from the ADIA Lab Causal Discovery Challenge organized by ADIA Lab and CrunchDAO [1, 3]. The full competition dataset contains approximately 47,000 synthetic graph instances generated from structural causal models with nonlinear mechanisms including neural networks, Gaussian processes, and Bayesian models [5]. The dataset is divided into three partitions accessed through the CrunchDAO platform. The training set contains 23,500 dataset instances with provided ground-truth adjacency matrices, used for model training and local validation. The public test set contains 1,880 dataset instances scored by the CrunchDAO platform during the competition, providing an intermediate evaluation checkpoint before final submission. The private test set is a held-out partition revealed only at final submission; the proposed model achieves 81.082% Balanced Accuracy on this partition, representing the headline result of this thesis.

4.1.2 Label Derivation from the Adjacency Matrix

The eight-class label for each variable $K \notin \{X, Y\}$ is determined by the four binary entries of the ground-truth adjacency matrix $\mathbf{A}$ corresponding to edges incident to K with respect to X and Y : $\mathbf{A}_{K,X}$ (presence of edge $K \rightarrow X$), $\mathbf{A}_{X,K}$ (presence of

edge $X \rightarrow K$), $\mathbf{A}_{K,Y}$, and $\mathbf{A}_{Y,K}$. Table 4.1 enumerates the mapping from these four bits to the eight causal role labels. A Confounder satisfies $K \rightarrow X$ and $K \rightarrow Y$, a Mediator satisfies $X \rightarrow K$ and $K \rightarrow Y$, a Collider satisfies $X \rightarrow K$ and $Y \rightarrow K$, and the remaining four roles capture asymmetric relationships where K involves exactly one of X or Y in a single directional edge. A detailed causal interpretation of each role is provided in Chapter 2 (Section 2.1.2).

Table 4.1: Label derivation from the adjacency matrix for variable K . Each label is determined by the four binary entries of $\mathbf{A}$ corresponding to edges incident to K with respect to X and Y . The first row for each label shows the four bits in the order $K \rightarrow X$, $X \rightarrow K$, $K \rightarrow Y$, $Y \rightarrow K$. All variables in the ADIA Lab dataset are normalized to $[-1, 1]$.

Label	$K \rightarrow X$	$X \rightarrow K$	$K \rightarrow Y$	$Y \rightarrow K$
Confounder	1	0	1	0
Mediator	0	1	1	0
Collider	0	1	0	1
Cause of X	1	0	0	0
Cause of Y	0	0	1	0
Consequence of X	0	1	0	0
Consequence of Y	0	0	0	1
Independent	0	0	0	0

4.1.3 Dataset Characteristics

Each dataset instance contains $n = 1000$ observations over $p \in \{3, \dots, 10\}$ variables, with X and Y pre-identified as the treatment and outcome pair and the edge $X \rightarrow Y$ guaranteed to be present in the ground-truth DAG. Instances with $p > 3$ contribute more than one variable K to the classification task, so the total number of (K, label) training pairs exceeds 23,500. The eight classes are not uniformly distributed: Independent is the most frequent class while Mediator and Collider have the fewest training samples. This class imbalance motivates both the inverse-frequency class weights used in the loss function (Section 3.7) and the use of Balanced Accuracy as the evaluation metric. The data are generated from six synthetic graph types and real-world food-web graphs, covering a range of causal structures that test the

model’s ability to distinguish fine-grained role differences [5].

4.2 Evaluation Metric: Balanced Accuracy

The evaluation metric for all experiments is Balanced Accuracy (BA), defined as the mean per-class recall across all $C = 8$ classes:

$$\text{BA} = \frac{1}{C} \sum_{c=1}^C \frac{\text{TP}_c}{\text{TP}_c + \text{FN}_c}. \quad (4.1)$$

Balanced Accuracy is the appropriate metric for three interconnected reasons. First, the class distribution is imbalanced with Independent as the majority class and Mediator and Collider as rare classes; reporting overall accuracy would overweight performance on the majority class and obscure failures on minority classes that are equally important for applications. Second, all eight causal roles carry equal weight in downstream decision-making – misclassifying a Mediator as a Consequence of X has consequences for intervention planning that are no less serious than misclassifying an Independent as a Confounder. Third, an uninformed model that always predicts the most frequent class achieves approximately $\frac{1}{8} = 12.5\%$ Balanced Accuracy, providing a clear floor for the scale of improvement needed. The competition organizers’ supervised baseline achieved approximately 58% while the first-place solution achieved 76.70% [5].

4.3 Experimental Baselines

4.3.1 External Baselines

Two external baselines establish the competitive landscape. The competition first-place solution [6] is the primary reference method, achieving 76.70% Balanced Accuracy on the public leaderboard. Its architecture consists of 3-channel edge tensor construction, a stem Conv1D layer, five residual Conv1D blocks, global average pooling, edge-type embedding merge, two self-attention layers, and dual heads (8-class node classification plus binary edge auxiliary). All improvements in this thesis are measured against a faithful reimplementaion achieving 73.96% local and 73.61% public leaderboard Balanced Accuracy, confirming reproducibility

of the deep learning approach and providing a rigorous baseline for measuring incremental contributions.

The machine learning baselines represent the feature engineering and gradient boosting paradigm pushed to its limit. v10 applies LightGBM to eleven hand-crafted statistical features including Pearson correlation, mutual information, and ANM scores, achieving 63.11%. v13 implements a comprehensive pipeline with over 300 engineered features combining correlation coefficients, partial correlations, conditional mutual information, HSIC scores, and outputs from four classical causal discovery algorithms (PC, LiNGAM, NOTEARS, GES), processed through an XGBoost/LightGBM/CatBoost ensemble with GNN refinement, achieving 72.64%. These results demonstrate that scalar feature engineering – regardless of the number of features or the sophistication of the ensemble – cannot approach the performance of the deep learning approach that preserves the full distributional shape of conditional relationships.

4.3.2 Internal Ablation Baselines

The internal ablation baselines track incremental contributions through the development chain. The starting point is v2, the faithful 3-channel reimplementation achieving 73.96% local and 73.61% public. The 8-channel extension without structural attention (v8b) isolates the contribution of multi-bandwidth kernel regression and ANM residuals, achieving 76.94%. Adding the structural attention bias (v11) raises this to 79.59%, isolating the architectural contribution of topology-aware attention. Combining with XY augmentation (v11+xyaug) reaches 81.14% local and 80.82% public, serving as the previous best result. The 2D pipeline variants isolate visual representation contributions: v26c (2D node images only, no edge pipeline) demonstrates the necessity of edge context by collapsing to 51.51%, while v26e base (full dual-pipeline without augmentation) achieves 80.47%, confirming that the 2D visual features provide complementary information that the 1D edge pipeline alone cannot capture.

4.4 Experimental Setup

4.4.1 Hardware Environment

All experiments are conducted on a server equipped with two NVIDIA RTX 5880 GPUs providing 48 GB combined VRAM, a 48-core CPU for parallelizing data preprocessing across multiple workers, and 251 GB of system RAM. Training uses Distributed Data Parallel across both GPUs to accelerate the training loop on the augmented dataset of approximately 263,000 samples. The model has approximately 280K parameters, comfortably fitting within the available GPU memory while allowing batch sizes of 64 for stable training.

4.4.2 Two-Tier Evaluation Procedure

The evaluation procedure follows a two-tier protocol with three distinct scoring mechanisms. The first tier is local validation: after each training run, the model is evaluated immediately on the 1,880 held-out instances from the public test split, which have known ground-truth adjacency matrices. This provides rapid feedback for architectural decisions without consuming submission quota on the CrunchDAO platform. The second tier is the CrunchDAO public leaderboard: after local validation confirms a positive result, selected model versions are submitted for official scoring, providing a fair comparison against other competition participants. The third mechanism is the private test submission: the final proposed model is submitted once to the private test partition, which remains hidden until the end of the competition, achieving 81.082% as the definitive performance estimate. This two-tier protocol (local development followed by platform submission) prevents overfitting to the public leaderboard while ensuring reproducibility through the local validation split.

4.4.3 Experiment Design Principles

Experiments follow an ablation-first protocol to ensure every architectural decision is grounded in evidence. For each model variant, training and evaluation proceed on the base 23,500-sample dataset first. If the local result shows clear improvement over the corresponding ablation baseline, the experiment is repeated with XY aug-

mentation (approximately 263K samples) before any public leaderboard submission. This ordering ensures that computational resources are allocated to configurations with demonstrated positive returns rather than being spent on speculative changes. The protocol also prevents overfitting to the public test set by limiting submissions to configurations that show unambiguous local improvement, a concern given that the public leaderboard represents only a fraction of the total test distribution.

4.5 Complete Experiment Summary

Table 4.2 presents all model versions with their local and public leaderboard scores, organized chronologically and grouped by experiment category. The table enables direct comparison of the contribution of each component: the gap between v2 (3-channel baseline) and v8b (8-channel) shows the effect of multi-bandwidth representation; the gap between v8b and v11 shows the effect of structural attention bias; the gap between v26e base and v26c shows the contribution of the edge context pipeline; and the gap between v26e base and v26e+xyaug shows the effect of X/Y remap augmentation. Detailed analysis of these results, including per-class breakdowns and mechanistic explanations, is presented in Chapter 5.

Chapter Summary

This chapter described the evaluation setup for assessing the proposed method. The dataset consists of 23,500 training instances and 1,880 local test instances from the ADIA Lab Causal Discovery Challenge, with a private test partition for final evaluation. Balanced Accuracy is the sole evaluation metric, defined as mean per-class recall across the eight causal roles, chosen because the class distribution is imbalanced and all eight roles carry equal decision-making weight. The baseline landscape spans two ML approaches (v10 at 63.11% and v13 at 72.64%), the reimplemented 3-channel deep learning baseline (v2 at 73.96% local, 73.61% public), the previous best configuration with structural attention bias (v11+xyaug at 81.14% local, 80.82% public), and the proposed final model (v26e 8ch + XY augmentation) achieving 81.082% on the private test set. Detailed analysis of these results, including per-class breakdowns, ablation interpretation, and mechanistic

explanations, is presented in Chapter 5.

Table 4.2: Complete experiment results. Local = 1,880-sample test set; LB = public leader-board; Private = hidden test (final only). $\Delta v2$ = change vs. v2 baseline (73.96%).

Version	Description	Local	LB	$\Delta v2$
<i>ML baselines</i>				
v10	LightGBM on 11 scalar features	63.11%	—	−10.85
v13	300+ features, 4 algo outputs	72.64%	—	−1.32
<i>DL reimplementations (3-channel baseline)</i>				
v2	Faithful 3ch reimplementations	73.96%	73.61%	—
<i>Multi-bandwidth extension (8-channel)</i>				
v5m	+multi-bw kernel (5ch)	75.44%	74.80%	+1.48
v5m+xyaug	v5m + XY aug	77.91%	77.55%	+3.95
v8b	8ch (5 kernel + 3 ANM)	76.94%	—	+2.98
v8b+xyaug	v8b + XY aug	80.26%	80.22%	+6.30
<i>Structural attention bias</i>				
v11	v8b + struct. bias	79.59%	—	+5.63
v11+xyaug	v11 + XY aug	81.14%	80.82%	+7.18
v11 ens	3-seed ensemble	80.92%	80.27%	+6.96
<i>2D scatter density pipeline</i>				
v26b	8ch pairwise Conv2D	79.17%	—	+5.21
v26c	8ch 2D only (no edge)	51.51%	—	−22.45
v26d	Edge loss favored ($\lambda = 1.0$)	80.37%	—	+6.41
v26d nf	Node loss favored ($\lambda = 0.3$)	78.77%	—	+4.81
v26e 12ch	12ch (incl. coeff. ch.)	80.42%	—	+6.46
v26e 8ch	8ch (coeff. removed)	80.47%	—	+6.51
<i>Final model (v26e 8ch + XY aug)</i>				
v26e+xyaug	Proposed	81.19%	80.96%	+7.23
v26e+xyaug	Private	—	—	81.082%

Chapter 5

Results and Analysis

This chapter presents the complete experimental results with mechanistic analysis of why each contribution improves performance and what general lessons can be extracted. Results are grouped by theme: Section 5.1 analyzes why deep learning outperforms machine learning baselines; Section 5.2 evaluates the multi-bandwidth and ANM channel contributions; Section 5.3 examines the structural attention bias; Section 5.4 presents the 2D pipeline ablations; Section 5.5 provides the cumulative ablation; and Section 5.6 synthesizes the general principles that emerged from this work.

5.1 Deep Learning vs Machine Learning

This section establishes the mechanistic reasons why deep learning outperforms machine learning on this task, with implications for how future causal discovery systems should be designed.

5.1.1 v10: LightGBM on Scalar Features

Version v10 trains a LightGBM classifier on eleven hand-crafted scalar features including Pearson correlation, Spearman correlation, mutual information, and ANM scores, achieving 63.11% Balanced Accuracy. This result is substantially below every Conv1D-based model in the thesis. The fundamental limitation is not the number of features or the choice of classifier but the irreversible nature of scalar compression: every feature in v10 reduces $n = 1000$ observations to a single number, discarding the functional form of the conditional relationship between variables.

Consider Mediator ($X \rightarrow K \rightarrow Y$) versus Consequence of X ($X \rightarrow K$ only). Both roles have the edge $X \rightarrow K$ in their DAG structure, creating dependence between X and K through the same mechanism. The distinguishing signal is whether K subsequently affects Y , which is reflected in the curvature, heteroscedasticity, and tail behavior of the regression curve $\hat{m}_{X \rightarrow K}$ and the chain regression $\hat{m}_{K \rightarrow Y}$. No

scalar summary of these curves – whether correlation, mutual information, or ANM score – can capture this discriminating information. The scalar encodes only the strength of the dependence, not its functional form.

5.1.2 v13: Full ML Fullstack Pipeline

Version v13 represents the maximum reasonable engineering effort under the ML paradigm: 300+ engineered features including correlation coefficients, partial correlations, conditional mutual information, HSIC, ANM scores, and outputs from four classical causal discovery algorithms (PC, LiNGAM, NOTEARS, GES), processed through an XGBoost/LightGBM/CatBoost ensemble with GNN refinement. This achieves 72.64% – below the 3-channel DL reimplement baseline (v2: 73.96%) and 8.44 percentage points below the proposed model.

The gap is mechanistic, not a matter of insufficient feature engineering. Every scalar feature in v13 is a fixed aggregation of the $n = 1000$ observations, and the information loss from this compression is irreversible. The spatial structure of the joint density, the shape of the conditional expectation function, the presence of multimodality or heteroscedasticity – all the information that distinguishes the eight causal roles – is discarded when observations are reduced to scalars. In contrast, the 8-channel edge tensor (channels 2–8) retains the full length- n curve, and Conv1D learns which aspects of that curve are discriminative. The functional form of the causal relationship, not its aggregate strength, is the discriminating signal, and it is the information that scalar compression irrevocably destroys.

Table 5.1 summarizes the comparison. Machine learning approaches have genuine advantages – interpretability, no GPU requirement, faster development cycles – that make them appropriate for many tasks. For this specific problem, however, the discriminating signal lives in the shape of nonlinear functional relationships rather than in aggregate statistics, making the information loss from scalar compression decisive.

5.2 Feature Engineering Contributions

This section analyzes the incremental contributions of each channel in the 8-channel edge tensor, establishing that multi-bandwidth kernel regression and ANM resid-

Table 5.1: Deep learning vs. machine learning on the ADIA Lab Challenge. Scalar compression (v13: 72.64%) is 8.55 points below the proposed model (v26e+xyaug: 81.19%) because the functional form of conditional relationships – the discriminating signal for the eight causal roles – is destroyed when $n = 1000$ observations are reduced to scalar aggregates.

Approach	Description	BA	Gap
ML minimal (v10)	11 scalar features	63.11%	−18.08
ML fullstack (v13)	300+ scalars + algo outputs	72.64%	−8.55
DL proposed (v26e+xyaug)	8-ch, $n = 1000$ curves	81.19%	—

uals each provide distinct and complementary improvements.

5.2.1 Multi-Bandwidth Kernel Regression Channels (+1.48%)

Extending the 3-channel baseline (v2: 73.96%) by adding multi-bandwidth kernel regression channels at $h \in \{0.2, 1.0\}$ alongside the original $h = 0.5$ produces v5m at 75.44%, a gain of +1.48%. Each bandwidth captures a different smoothness scale of the nonlinear conditional expectation: $h = 0.2$ is sensitive to sharp local transitions and fine nonlinear structure, while $h = 1.0$ reflects global monotone trends and is robust to local noise. Using all three bandwidths together prevents the model from depending on any single fixed choice of h , which is critical because the optimal smoothness varies across different causal mechanisms and data generating processes. The gain is confirmed on the public leaderboard: v5m+xyaug achieves 74.80% LB, a +1.97% improvement over the corresponding baseline, confirming that the improvement reflects genuine generalization rather than overfitting to the local test split.

The multi-bandwidth channels address a fundamental limitation of single-bandwidth representations: no fixed bandwidth is optimal for all causal relationships. Some variable pairs have sharp threshold-like transitions (requiring small h), while others have smooth monotone relationships (better captured by large h). The ANM residual channels, in particular, benefit from multiple bandwidths because the directional asymmetry signal is present at different scales depending on the functional form of the structural equations.

5.2.2 ANM Residual Channels (+1.50%)

Adding three ANM residual channels to produce the complete 8-channel tensor (v8b: 76.94%) improves over the 5-channel version (v5m: 75.44%) by +1.50%. The improvement is concentrated in Collider (+6.4%) and Consequence of X (+5.0%), classes where detecting the directionality of the causal edge is the primary discriminating challenge. The ANM residual $r_{u \rightarrow v} = v - \hat{m}(u)$ is approximately independent of u when $u \rightarrow v$ is the true causal direction, producing a flat residual curve. Under the reversed direction, the residual retains systematic nonlinear structure. This directional asymmetry – flat versus structured – is visible in the 1D residual curve and is what the Conv1D network learns to detect. The multi-bandwidth formulation ensures this directional signal is captured at multiple smoothness scales, making the ANM channel contribution robust across different functional forms.

5.3 Structural Attention Bias

This section analyzes the single most impactful architectural change: the topology-aware attention mechanism that injects causal structural priors using only six learned scalars.

5.3.1 Result and Mechanism (+2.65%)

Adding structural attention bias to v8b (v11: 79.59% vs. v8b: 76.94%) yields +2.65%, the largest gain from any single architectural modification across all 30+ experiments conducted in this thesis. The gain is distributed across classes, with the largest improvements in Confounder (+4.46%), Collider (+3.40%), and Mediator (+1.04%). These three classes are precisely the ones whose definitions involve triangle motifs – chain ($X \rightarrow K \rightarrow Y$), fork ($K \rightarrow X, K \rightarrow Y$), and collider ($X \rightarrow K \leftarrow Y$) – that are directly encoded in the six topological relationship types used in the structural attention bias. This distributed gain confirms that the bias is acting on the correct structural feature.

The mechanism operates during attention aggregation: edges that share an endpoint or form a directed chain through a node of interest receive differential attention weights through the learned scalar bias, rather than being treated identically

to unrelated background edges. For example, in a Mediator chain $X \rightarrow K \rightarrow Y$, the edges $X \rightarrow K$ and $K \rightarrow Y$ form a forward chain relationship (Type 4 in Table 3.1) and receive a shared bias that strengthens their mutual attention, propagating causal context along the pathway.

5.3.2 Why Structured Inductive Bias Outperforms Unconstrained Capacity

Versions v3, v4, v6, v9, and v9b all attempt architectural additions that add unconstrained parameters – parallel observation cross-attention branches, dual-path transformers, observation-level transformers, node-centric aggregation variants – and all produce results at or below the 3-channel baseline. The structural attention bias (v11, +2.65%) is the only successful architectural modification among all these experiments.

The distinguishing factor is not parameter count but the alignment of inductive bias with domain knowledge. The six topological relationship types encode explicit prior knowledge about causal motifs (chain, fork, collider, reverse) that the model should attend to. Six additional scalars with this structured prior outperform thousands of additional unconstrained parameters by factors exceeding 3.5 percentage points. This validates the principle that for this problem, structured inductive bias consistently outperforms increased unconstrained capacity. The structured prior does not learn a new representation from scratch; it only learns the relative priority among topologically distinct edge pairs, which is precisely the information encoded in the causal DAG structure.

The lesson extends beyond this specific task: when the domain has known structural constraints (as causal graphs do), encoding those constraints as architectural priors with minimal parameter cost is more effective than adding representational capacity and hoping the model discovers those constraints from data alone.

5.4 2D Scatter Density Pipeline

This section presents the ablation studies for the 2D scatter density pipeline, establishing that the edge context is structurally necessary, the 1D and 2D representations are complementary, and the final model achieves 81.19% local and 81.082%

private Balanced Accuracy.

5.4.1 Edge Context Is Structurally Necessary

Table 5.2 presents the ablation evidence for the necessity of edge context in the dual-pipeline architecture. When the entire 1D edge pipeline is removed and nodes are classified from their 2D visual embeddings alone (v26c), Balanced Accuracy collapses to 51.51% – a 28.96-point drop from the full model’s 80.47%. A variant using mean pooling of all node embeddings as a substitute for edge context (v28b) achieves only 46.39%. These results confirm that the 2D density image captures only the local distributional structure of K relative to X and Y but cannot reason about K ’s causal role within the larger graph structure. The edge encoder, supervised by the binary edge loss against ground-truth adjacency matrices, provides exactly this graph-wide structural context through the four attended edge embeddings ($K \rightarrow X$, $K \rightarrow Y$, $X \rightarrow K$, $Y \rightarrow K$). Without it, performance is worse than the ML baseline, confirming that the edge head is a structural necessity rather than auxiliary regularization.

Table 5.2: Ablation on the necessity of edge context for node classification. All three variants use the same 8-channel 32×32 node images; only the source of graph-wide structural context differs. The full model (v26e 8ch) combines edge context with visual features, achieving 80.47%. Removing the edge pipeline entirely (v26c) collapses performance by 28.96 points to 51.51%, worse than the ML baseline. Replacing edge context with mean-pooled node embeddings (v28b) further degrades to 46.39%. The binary edge loss, which supervises the edge embeddings against ground-truth adjacency entries, is essential for maintaining causally well-structured representations that the node merger depends on.

Version	Graph Context Source	Local BA
v26c	None (pure 2D visual, no edge pipeline)	51.51%
v28b	Mean pooling of all node embeddings	46.39%
v26e 8ch (full model)	Edge encoder + structural self-attention	80.47%

5.4.2 1D and 2D Representations Are Complementary

Table 5.3 shows the per-class breakdown comparing the 1D edge-only model (v11: 79.59%) with the full dual-pipeline model (v26e 8ch: 80.47%). The 2D pipeline gains most on Confounder (+2.78%), Mediator (+2.28%), and Consequence of Y (+2.18%) – classes whose distinguishing feature is the joint distributional pattern of K with respect to both X and Y simultaneously. The 1D edge tensor captures each pair separately as marginal curves; the 2D image captures the joint view of K against X and Y together, revealing patterns such as correlated variation across both conditioning variables that are invisible in any single 1D curve. Collider drops slightly (−2.02%), suggesting that the 1D ANM residual channels – which detect directional asymmetry in marginal pairwise relationships – are more powerful than 2D density for detecting the collider signature (induced dependence under conditioning). This class-level complementarity confirms that the two representations capture structurally different information about each node’s causal role.

5.4.3 Loss Lambda Ablation

Table 5.4 shows the effect of the edge loss weight λ on the final model (v26e 8ch, no XY augmentation). The configuration with $\lambda = 0.7$ (80.47%) outperforms both $\lambda = 0.3$ (node-favored, 78.77%) and $\lambda = 1.0$ (edge-only, 80.37%).

The large gap between $\lambda = 0.3$ and $\lambda = 0.7$ (−1.70%) reflects a training dynamics effect: when the node loss gradient dominates early in training, the edge pipeline receives insufficient supervision and produces inconsistent edge embeddings that degrade the quality of the node merger. The edge pipeline must remain well-supervised throughout training because the node classifier depends entirely on the edge embeddings for graph-wide structural context. The small gap between $\lambda = 1.0$ and $\lambda = 0.7$ (−0.10%) indicates that any configuration with sufficiently high edge weight is adequate; $\lambda = 0.7$ is chosen as it empirically performs best.

5.4.4 Final Model Per-Class Performance

Table 5.5 presents the per-class recall of the final proposed model (v26e 8ch + XY augmentation) on the local test set. The three hardest classes – Confounder (74.93%), Mediator (73.71%), and Collider (78.70%) – remain below the overall av-

Table 5.3: Per-class recall comparison between the 1D edge-only model (v11, 79.59%) and the dual-pipeline model (v26e 8ch, 80.47%), evaluated on the base dataset without XY augmentation. The 2D pipeline provides the largest gains on Confounder (+2.78%), Mediator (+2.28%), and Consequence of Y (+2.18%), where joint distributional patterns with both X and Y are more informative than marginal pairwise curves. The 1D pipeline outperforms 2D on Collider (-2.02%) because the ANM residual channels capture directional asymmetry more effectively than 2D density for this class, which is distinguished by conditional induction of dependence rather than joint distributional structure. Neither representation fully dominates the other, confirming the complementary nature of 1D pairwise and 2D joint representations.

Class	v11 (1D only)	v26e 8ch (1D+2D)
Confounder	75.81%	78.59%
Collider	77.80%	75.78%
Mediator	67.29%	69.57%
Cause of X	82.91%	82.40%
Cause of Y	83.54%	84.71%
Consequence of X	79.69%	80.20%
Consequence of Y	81.96%	84.14%
Independent	87.67%	88.38%
Balanced Accuracy	79.59%	80.47%

erage, consistent with competition-wide findings [5]. These are precisely the classes where pairs of roles share similar conditional independence signatures (Confounder vs Cause of X; Mediator vs Consequence of X; Mediator vs Confounder) and where scalar features are most inadequate. The per-class improvements over the reimplement baseline (v2: 73.96%) are largest for Mediator (+13.7%), Collider (+12.7%), and Consequence of X (+8.6%), confirming that the 8-channel tensor and structural attention bias specifically address the mechanistic challenges that make these classes difficult.

Table 5.4: Effect of loss weight λ on v26e 8ch (base, no XY augmentation). The total loss is $\mathcal{L} = \lambda \cdot \mathcal{L}_{\text{edge}} + (1 - \lambda) \cdot \mathcal{L}_{\text{node}}$. Node-favored training ($\lambda = 0.3$) produces degenerate edge embeddings that degrade node merger quality, causing a 1.70-point drop. Edge-only training ($\lambda = 1.0$) slightly underperforms $\lambda = 0.7$ by 0.10 points, suggesting some node gradient is beneficial for stabilizing the fusion layer. The optimal $\lambda = 0.7$ reflects the need to keep the edge pipeline well-supervised throughout training, as the node classifier depends entirely on edge embeddings for graph-wide structural context.

Configuration	λ	Local BA
v26d node-favored	0.3	78.77%
v26d edge=node	1.0	80.37%
v26e (proposed)	0.7	80.47%

5.5 Incremental Ablation

Table 5.6 summarizes the cumulative contribution of each proposed component as incremental gains over the baseline, enabling direct comparison of the relative impact of each contribution.

5.6 Synthesis and Lessons Learned

This section extracts four general principles from the experimental results that extend beyond the specific ADIA Lab task to the broader problem of causal discovery from observational data.

5.6.1 Shape Information Cannot Be Recovered After Scalar Compression

The v13 ML fullstack experiment establishes this as the central finding of the thesis. 300+ features, four causal discovery algorithms, and three gradient boosting models together achieve 72.64% – lower than every Conv1D-based model that operates on even the minimal 3-channel functional curve representation. The functional form of the causal relationship – its curvature, multimodality, heteroscedasticity, tail behavior, and directional asymmetry – contains discriminative information that is not accessible through any scalar summary. Once observations are reduced to ag-

Table 5.5: Per-class recall and sample counts for the final proposed model (v26e 8ch + XY augmentation) on the 1,880-instance local test set. The three hardest classes – Mediator (73.71%), Confounder (74.93%), and Collider (78.70%) – are precisely the classes where role pairs share similar conditional independence signatures and require graph-wide structural context to resolve. Independent (88.38%) is the easiest class because its defining property (no edges to X or Y) is detectable from marginal independence alone. The final model achieves 81.19% Balanced Accuracy locally, 80.96% on the public leaderboard, and 81.082% on the private test set.

Class	Recall	Count (n)
Confounder	74.93%	682
Collider	78.70%	446
Mediator	73.71%	483
Cause of X	83.32%	983
Cause of Y	85.45%	2145
Consequence of X	81.29%	1571
Consequence of Y	83.72%	682
Independent	88.38%	4494
Balanced Accuracy	81.19%	

gregate statistics, this information is irrevocably lost. This has direct implications for the design of future causal discovery systems: when the task requires distinguishing roles that differ in the shape of conditional relationships, scalar feature engineering is insufficient regardless of the number of features or the sophistication of the classifier.

5.6.2 Structured Inductive Bias Consistently Outperforms Unconstrained Capacity

The structural attention bias (v11, +2.65%) is the only successful architectural modification among all 30+ experiments conducted in this thesis. Every other architectural addition – parallel observation branches, dual-path transformers, observation-level transformers, node-centric aggregation, deep sets – produces results at or below the baseline. The distinguishing factor is not parameter count but the align-

Table 5.6: Incremental ablation: cumulative Balanced Accuracy gains from baseline (v2: 73.96%) to final model (81.19%, +7.23 points). Bottom section: 2D pipeline contribution (+0.88%) measured against 1D-only v11 base, not directly incremental to the v11 row above.

Configuration	Description	Local BA	Δ
v2: baseline	3ch reimplementation	73.96%	—
v5m	+multi-bw kernel (5ch)	75.44%	+1.48
v8b	+ANM residual ch. (8ch)	76.94%	+1.50
v11	+structural attention bias	79.59%	+2.65
v11+xyaug	+XY augmentation	81.14%	+1.55
v26e 8ch (1D+2D)	+2D scatter density	80.47%	+0.88*
v26e+xyaug	final model	81.19%	+0.72

*v26e base vs. v11 base: 1D+2D vs. 1D-only (+0.88%).

ment of inductive bias with domain knowledge. Six learnable scalars encoding six topological relationship types outperform thousands of unconstrained parameters by over 3.5 percentage points. This validates the principle that for problems with known structural constraints, encoding those constraints as architectural priors with minimal parameter cost is more effective than adding representational capacity and hoping the model discovers the constraints from data alone. The structured prior does not learn a new representation – it only learns the relative priority among topologically distinct edge pairs.

5.6.3 1D Curves and 2D Density Images Capture Complementary Information

The 1D edge tensor representation captures directional pairwise relationships between variable pairs as marginal curves, encoding causal asymmetry through ANM residuals. The 2D density image captures the joint distributional pattern of a node with respect to both X and Y simultaneously, encoding joint variation patterns that are invisible in any single 1D marginal curve. Neither representation fully subsumes the other: the 1D pipeline is stronger for Collider (where directional residual asymmetry is the discriminating signal), while the 2D pipeline is stronger for Con-

founder, Mediator, and Consequence of Y (where joint distributional structure with both X and Y distinguishes the roles). The fusion of both representations (v26e: 80.47%) outperforms either alone (v11 1D-only: 79.59%; v26c 2D-only: 51.51%).

5.6.4 Edge Context Is the Structural Foundation, Not an Auxiliary

Despite the “auxiliary” label attached to the binary edge head, the edge pipeline is the structural foundation of the entire model. The 28.96-point collapse of v26c (51.51%) when the edge pipeline is removed demonstrates that no amount of visual information in the 2D density images can compensate for the absence of graph-wide context. The edge encoder, supervised by the binary edge loss against ground-truth adjacency entries, forces the edge embeddings to encode causally meaningful structure – specifically, whether each directed edge exists in the ground-truth DAG – that the node merger requires to determine each node’s role. The binary edge loss is not regularization; it is essential supervised context encoding that maintains the discriminative quality of edge embeddings throughout training. Removing this supervision produces degenerate edge representations that degrade the node merger regardless of the quality of the node images.

Chapter Summary

This chapter presented the complete experimental analysis of the proposed method. Six key findings emerged from this work. First, scalar feature compression is irreversible and structurally insufficient for causal role classification: even 300+ features and three ensemble models achieve only 72.64%, below every Conv1D model. Second, multi-bandwidth kernel regression and ANM residual channels each provide consistent approximately +1.5% improvements through different mechanisms – smoothness scale and directional asymmetry respectively. Third, structural attention bias provides the largest single architectural gain of +2.65% through six learnable scalars encoding topology-aware priors, validating that structured inductive bias outperforms unconstrained capacity by over 3.5 percentage points. Fourth, the 2D scatter density pipeline contributes +0.88% with complementary class-level improvements, confirming that 1D pairwise and 2D joint representations capture different information. Fifth, edge context is structurally necessary: removing it col-

lapses performance by 29 points, confirming the edge loss is essential supervision rather than auxiliary regularization. Sixth, XY augmentation provides +0.72% to +3.32% depending on base model strength, with diminishing returns as the base model becomes richer. The final proposed model achieves 81.19% locally, 80.96% on the public leaderboard, and 81.082% on the private test set, surpassing the competition first-place solution (76.70%) by +4.38% and the reimplementation baseline (73.96%) by +7.12%.

Conclusion

This thesis addressed the challenge of classifying causal variable roles in observational data, focusing on the ADIA Lab Causal Discovery Challenge where the goal is to assign one of eight labels (Confounder, Mediator, Collider, Independent, Cause of X/Y, Consequence of X/Y) to variables given their relationships with designated treatment X and outcome Y . The problem is difficult because several role pairs share nearly identical statistical signatures – for example, Confounder and Mediator both produce conditional independence $X \perp Y \mid K$ – requiring models to distinguish subtle structural patterns rather than merely testing for dependence.

The proposed solution is a **dual-pipeline neural architecture** combining 1D edge tensor processing with 2D scatter density visualization, achieving **81.082%** Balanced Accuracy on the private test set. This represents a **+4.38% improvement** over the competition first-place solution (76.70%) and a **+7.12% improvement** over a faithful reimplement baseline (73.96%). The model makes five specific technical contributions, each validated by systematic ablation: (1) 8-channel edge tensors with multi-bandwidth kernel regression and ANM residuals (+1.89%); (2) structural attention bias encoding six topological relationship types (+2.65%); (3) 2D scatter density pipeline capturing joint distributional patterns (+0.88%); (4) λ -weighted dual-loss training ensuring edge context quality (+0.59%); and (5) X/Y remap data augmentation multiplying effective training data 11-fold (+1.55%).

Beyond the numerical improvements, this thesis establishes four methodological insights with broader relevance to causal discovery systems. **First**, scalar feature compression is irreversible – once the full shape of conditional relationships is reduced to correlation coefficients or independence test statistics, the information distinguishing Confounder from Mediator is permanently lost. The 8-channel edge tensor preserves the complete functional curve, enabling Conv1D to learn discriminative patterns invisible to scalar methods. **Second**, structured inductive bias consistently outperforms unconstrained capacity. The structural attention bias requires only six additional parameters yet provides +2.65% gain, while failed experiments adding thousands of unconstrained parameters (parallel transformers, node-centric attention, deep sets) produced negative gains of -1.19% to -1.90%. **Third**, 1D curve representations and 2D density images are complementary. Edge

tensors excel at directional asymmetry (Collider detection), while scatter density images capture joint patterns (Confounder/Mediator distinction); neither representation subsumes the other. **Fourth**, edge context is a structural necessity, not an auxiliary regularization term. Removing the edge pipeline causes performance to collapse from 80.47% to 51.51% – below even classical machine learning baselines – confirming that the binary edge loss forces embeddings to encode causally meaningful graph structure essential for node classification.

Limitations and future work. Despite achieving state-of-the-art performance, three challenges remain. Mediator (73.71%) and Collider (78.70%) remain the hardest classes with substantial room for improvement. The model has been validated only on synthetic data from the ADIA Lab competition; real-world causal graphs may exhibit different noise characteristics, missing data patterns, or temporal dynamics not present in the training distribution. Finally, the architecture requires GPU resources for both training and inference, limiting deployment in resource-constrained environments. Future work should validate the approach on real-world causal inference benchmarks, extend the method to time series causal discovery where temporal ordering provides additional constraints, and investigate distillation techniques to reduce model size while preserving accuracy.

This work demonstrates that deep learning architectures can advance causal discovery beyond the limitations of classical methods when design choices are grounded in the structural properties of causal graphs. The 81.082% Balanced Accuracy establishes a new state-of-the-art for the ADIA Lab Challenge and provides a foundation for automated causal analysis systems that can reason about variable roles in complex systems.

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